ResearchGate

See discussions, stats, and author profiles for this publication at: https://www.researchgate.net/publication/258756978

The Role of Fluids in Tectonic and Earthquake Processes

Article *in* Advances in Geophysics · December 2013

DOI: 10.1016/B978-0-12-380940-7.00001-9

CITATIONS
99

READS
3,329

1 author:

Stephen Miller
University of Neuchâtel

110 PUBLICATIONS **3,781** CITATIONS

SEE PROFILE

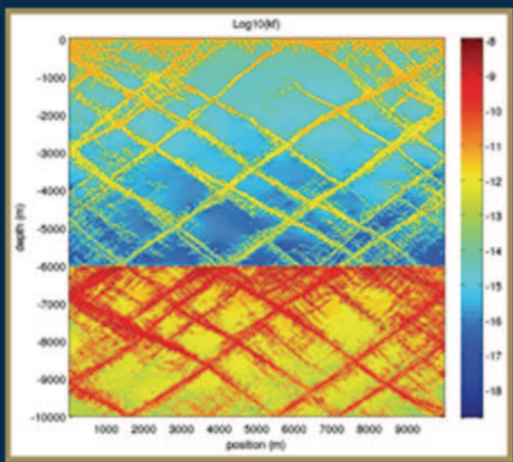
All content following this page was uploaded by Stephen Miller on 24 March 2018.

The user has requested enhancement of the downloaded file.

Advances in

GEOPHYSICS

VOLUME 54



VOLUME FIFTY FOUR

ADVANCES IN GEOPHYSICS

This page intentionally left blank

VOLUME FIFTY FOUR

ADVANCES IN GEOPHYSICS

Series Editor

RENATA DMOWSKA

School of Engineering and Applied Sciences,
Harvard University, Cambridge,
Massachusetts, USA

ELSEVIER

Amsterdam • Boston • Heidelberg • London
New York • Oxford • Paris • San Diego
San Francisco • Singapore • Sydney • Tokyo
Academic Press is an imprint of Elsevier

Academic Press is an imprint of Elsevier
525 B Street, Suite 1800, San Diego, CA 92101-4495, USA
225 Wyman Street, Waltham, MA 02451, USA
32 Jamestown Road, London NW1 7BY, UK
The Boulevard, Langford Lane, Kidlington, Oxford, OX5 1GB, UK
Radarweg 29, PO Box 211, 1000 AE Amsterdam, The Netherlands

First Edition 2013

Copyright © 2013 Elsevier Inc. All rights reserved.

No part of this publication may be reproduced, stored in a retrieval system or transmitted in any form or by any means electronic, mechanical, photocopying, recording or otherwise without the prior written permission of the publisher.

Permissions may be sought directly from Elseviers Science & Technology Rights Department in Oxford, UK: phone (+44) (0) 1865 843830; fax (+44) (0) 1865 853333; email: permissions@elsevier.com. Alternatively, you can submit your request online by visiting the Elsevier web site at http://elsevier.com/locate/permissions, and selecting Obtaining permission to use Elsevier material.

Notices
No responsibility is assumed by the publisher for any injury and/or damage to persons or property as a matter of products liability, negligence or otherwise, or from any use or operation of any methods, products, instructions or ideas contained in the material herein. Because of rapid advances in the medical sciences, in particular, independent verification of diagnoses and drug dosages should be made.

ISBN: 978-0-12-380940-7
ISSN: 0065-2687

For information on all Academic Press publications
visit our website at store.elsevier.com

Printed and bound in United States of America

13 14 15 10 9 8 7 6 5 4 3 2 1

Working together to grow
libraries in developing countries
www.elsevier.com | www.bookaid.org | www.sabre.org
ELSEVIER BOOK AID International Sabre Foundation

CONTENTS

This page intentionally left blank

CONTRIBUTORS

Delphine Croizé
Department of Geosciences, University of Oslo, Oslo, Norway.

Dag K. Dysthe
Physics of Geological Processes, University of Oslo, Norway; Laboratoire Interdisciplinaire de Physique, Grenoble, France.

Jean-Pierre Gratier
Institut des Sciences de la Terre, Observatoire, CNRS, Université Joseph Fourier-Grenoble I, Grenoble, France.

Stephen A. Miller
Geodynamics and Geophysics, Steinmann Institute, University of Bonn, Bonn, Germany.

François Renard
Institut des Sciences de la Terre, Observatoire, CNRS, Université Joseph Fourier-Grenoble I, Grenoble, France; Physics of Geological Processes, University of Oslo, Norway.

This page intentionally left blank

CHAPTER ONE

The Role of Fluids in Tectonic and Earthquake Processes

Stephen A. Miller
Geodynamics and Geophysics, Steinmann Institute, University of Bonn, Bonn, Germany

Contents

Advances in Geophysics, Volume 54
ISSN 0065-2687, http://dx.doi.org/10.1016/B978-0-12-380940-7.00001-9
© 2013 Elsevier Inc.
All rights reserved.

Abstract

Fluids play an integral role in the geodynamical system, from consumption through serpentinization at mid-ocean ridges and outer rises, to release through dehydration and decarbonization within subduction zones and beyond. Fluids affect a number of critical elements of the tectonic cycle, including weakening plate boundaries and catalyzing mantle wedge melting for feeding volcanic arcs. This review paper summarizes the vast topic of the hydrogeological cycle of the solid earth, and how fluids affect, and are affected by, tectonic processes. Ultimately these fluids must either remain trapped in the mantle or return to the surface at high pressure via ductile processes or fracture networks. High pressure fluids returning to the surface may get trapped at the base of the brittle crust, where they can contribute to earthquake nucleation and genesis. Evidence suggests that high pressure fluids are active participants in tectonic earthquakes, and the relatively recent discovery of slow slip earthquakes and non-volcanic tremor phenomena all point to trapped, over-pressured fluids as an underlying mechanical cause. Fluids play an integral role in lithospheric geodynamics, which provides for some speculations about fluids and earthquakes in a general sense. One such speculation is that spatial aftershock patterns reflect fluid pathways taken by the release of high pressure fluids triggered by the earthquake mainshock. Some of these patterns are shown, and I introduce the term "Zen Trees" to describe them because of their aesthetic form and their resemblance to Eastern calligraphy. I hypothesize that earthquakes that do not spawn significant aftershock sequences indicate little if any trapped high pressure fluids at depth, while earthquakes producing long-lived aftershock sequences point to large reservoirs of trapped high pressure fluids. Although the viscous mantle is the ultimate geophysical fluid, the focus in this paper is limited to fluids in the lithosphere because this boundary, typically treated as a thermal boundary layer, is controlled by complex dynamical interactions between fracture, deformation, dissolution/precipitation, and fluid flow.

Nothing is weaker than water
But for attacking what is tough
Nothing surpasses it
Nothing equals it
Lao Tzu, Tao Te King, ca. 500 BC.

1. INTRODUCTION

The Chinese philosopher Lao Tzu knew nothing about tectonics or geodynamics, but he correctly identified the dominant role that fluids play in nature; nothing being weaker, yet nothing more capable of bringing down a mountain. Over two thousand years later, Terzaghi recognized that fluid pressure can completely neutralize the strengthening effect of normal stress in frictional materials (Terzaghi, 1923), a recognition now commonly known as Terzaghi's effective stress principle. The implication of the effective stress law (Nur & Byerlee, 1971) is that the lithosphere has zero shear strength when the pressure of the fluid approaches the rock pressure. Byerlee's Law (Byerlee, 1978), applicable to most rock types in the brittle field, dictates substantial strength of the upper lithosphere under hydrostatic fluid pressure. However, vast regions within the lithosphere, particularly along plate boundaries, host fluids without hydraulic connectivity to the free-surface. With no free-surface connection, fluid is likely overpressured, limited by the minimum principle stress (plus a nominal tensile strength) after which tensile fracture occurs in the direction of the maximum principle stress. These trapped, over-pressured reservoirs have limited shear strength, and thus are the weakest link in the tectonic chain. Understanding the spatio-temporal evolution of these over-pressured reservoirs is complicated by the large-scale changes in hydraulic properties over short timescales concomitant with fracture or faulting. Meanwhile, free fluid phases that are precipitated and stored as hydrous minerals can clog hydraulic pathways, while volatiles currently trapped can be released during dehydration and decarbonization. The large scale hydraulic property changes occur precisely when fluids are consumed (permeability shutdown) (Jamtveit, 2010), and when fluids are released (permeability enhancement) (Micklethwaite & Cox, 2004; Miller, van der Zee, Olgaard, & Connolly, 2003).

Mantle degassing, a process nearly as old as Earth, continues unabated (McLean, 1985; Ingebritsen & Manning, 2002). The pathway taken by the fluids would be those places most tectonically active, e.g. volcanic arcs and plate boundaries (Irwin & Barnes, 1980), because this is where large-scale fracturing (e.g. earthquakes) and most devolatilization reactions occur. Low permeability in tectonically static regions, on the other hand, limits significant fluid flow. The importance of fluids in tectonic processes has long been recognized (Hill, 1977; Hubbert & Rubey, 1959; Nur & Booker, 1972; Sibson, 2000), but an appreciation of just how important fluids are has accelerated in recent years with the discovery of episodic tremor and slow slip earthquakes (ETS) (Rogers and Dragert, 2003; Shelly et al., 2007). ETS phenomena, found along subducting interfaces (Gomberg and Cascadia Working Group, 2010) and within the down-dip extensions of plate bounding faults (Becken, Ritter, Bedrosian, & Weckmann, 2011; Becken et al., 2008), leads to circumstantial evidence of near lithostatic fluid pressures trapped at depth (Thomas, Nadeau, & Burgmann, 2009). Although the fundamental underlying mechanism driving ETS is not yet known, ETS identifies the likelihood of high pressure fluids trapped at the base of locked seismogenic zones, thus implicating high pressure fluids as a potentially integral component of the earthquake nucleation process.

The hypothesis that high pressure fluids act as an important tectonic driver suggests that lithospheric deformation, and the earthquake cycle, is substantially controlled by the mechanical, chemical, and time-varying hydraulic behavior of high pressure fluids trapped at depth. Understanding the role of fluids in tectonic processes draws from a large range of disciplines, and in this paper, I attempt to distill these disparate fields into a coherent narrative of the hydrogeologic cycle of the solid earth and its influence on the tectonic system. Quantification of the total global water budget is discussed elsewhere (Bodnar et al., 2013).

The structure of this paper follows fluids from their consumption at mid-ocean ridges to their ultimate release along plate boundaries, through volcanic arcs, or transport deeper into the mantle. The influence of fluids on tectonic processes is discussed within the context of this cycle, with an emphasis on the role of fluids on the earthquake cycle. I will make the case that abundant fluids are trapped at high pressure within the lithosphere and at the base of seismogenic zones, and that these fluids escape primarily by large-scale and transient changes in permeability accompanying local fracture and crustal scale earthquakes. If earthquakes provide a trigger for fluid escape, then the pathway taken by the fluids will influence aftershock sequences because the high fluid pressure reduces the effective normal stress while fluid pressure gradients introduce pore-elastic stresses on the system (Rozhko, 2010). I present some aftershock sequences that appear as branching structures ("Zen Trees"), with consecutive aftershock hypocenters supplying the "paint and brush." Since it appears that many aftershock sequences, independent of their mother earthquake, produce these structures, then many aftershock sequences may be driven in large part by the same mechanism of degassing of trapped high pressure fluids at depth. Conversely, earthquakes that spawn few, if any aftershocks, would imply that no high pressure reservoir exists at depth, and aftershocks quickly die out after the short timescale readjustment of dislocation-induced stress changes.

Future research directions are also proposed, with a discussion about promising numerical modeling strategies for understanding these complex systems. I attempt to cover most aspects of the role of fluids in tectonic processes, and although a complete treatment is not possible in a single paper, I address each of the topics and steer the interested reader to the relevant works.

2. FLUID FLUID IN POROUS AND FRACTURED NETWORKS

Throughout the tectonic cycle, fluids are consumed via metasomatism and created in metamorphic devolatilization reactions, traveling as free fluids through fractured or porous networks, or via coupled ductile deformation and fluid flow. Fluids facilitate melting, weaken plate interfaces, or remain trapped in high pressure reservoirs waiting for their turn to flow. The timescales for these processes depend predominantly on the intrinsic permeability of the matrix, where permeability is a measure of the ability to transmit fluids through a solid porous skeleton or fracture network. Permeability (measured in m^2) is an elusive parameter because it can change by many orders of magnitude over short (and transient) timescales (Miller & Nur, 2000).

Fluid transport in the brittle lithosphere is controlled by flow through porous or interconnected fracture networks, described by Darcy's Law, which can be written:

$$v_x = -\frac{k}{\eta}\left(\frac{\partial P}{\partial x}\right); \quad v_y = -\frac{k}{\eta}\left(\frac{\partial P}{\partial y}\right); \quad v_z = -\frac{k}{\eta}\left(\frac{\partial P}{\partial z} - g\rho_{\text{fluid}}\right), \tag{1.1}$$

where v is the Darcy velocity in the x, y, and z-directions, P is the fluid pressure, g is the gravitational constant, ρ is density, k is the permeability, and η is the dynamic fluid viscosity. Higher flow rates occur from increasing pressure gradients, lowering fluid viscosity, or increasing the permeability. Although variations can exist for all terms in Eq. (1.1), only the permeability has the capacity to change many orders of magnitude over short time and space scales in response to typical tectonic processes. Many studies aimed at constraining k in the lithosphere (Ingebritsen & Manning, 2010) find crustal values of $k \approx 10^{-16}$–10^{-18} m^2, but these are geologically averaged values which, although probably accurate on average, smooth out the on—off behavior of permeability where most of the action takes place.

Fluid pressure diffuses in a porous medium by

$$\frac{\partial P}{\partial t} = \frac{k}{\eta\phi\beta}\nabla^2 P + \frac{\dot{\Gamma}}{\phi\beta}, \tag{1.2}$$

where ϕ is the porosity and β is the lumped compressibility of the fluid and pore space (Segall & Rice, 1995). The term, $\dot{\Gamma}$ represents a source term, such as would be needed for including fluid created (source) in metamorphic dehydration or decarbonization reactions, or consumed (sink) or metasomatic reactions. Equation (1.2) is equivalent to the heat conduction equation with a source term, where the first term on the right hand side is diffusion, and the second term the source. As with Eq. (1.1), pressure diffusion is controlled predominantly by the permeability.

2.1 Fracture Flow

Fractures are most efficient for transporting fluids because fracture permeability, in its simplest form assuming laminar flow between parallel fracture walls, is

$$k = \frac{b^2}{12}, \tag{1.3}$$

where b is the crack aperture measured in meters. More complicated forms that consider permeable walls (Mohais, Xu, Dowd, & Hand, 2012) or that consider tortuous paths around asperities can modify Eq. (1.3), but the b^2 term remains and is the dominant property controlling fluid flow. Such a drastic effect of crack aperture on permeability results in a wide range of values, for example, from $k \approx 10^{-13}$ m^2 for a crack aperture of 1 μm to 10^{-9} m^2 for crack apertures of 0.1 mm. Importantly, this is four orders of magnitude change in flow rates and diffusion timescales for very small (and geologically relevant) changes in crack apertures. An excellent example of the complexity of permeability comes from a variety of measurements of basaltic crust (Fisher, 1998), where permeability takes on a spectrum of values depending on the measurement technique; from $k \approx 10^{-22}$–10^{-17} m^2 for intact specimens, $k \approx 10^{-18}$–10^{-13} m^2 in situ, and $k \approx 10^{-16}$–10^{-9} m^2 from temperature and flow meter logs. In another study from the East Pacific Rise (Crone, Tolstoy, & Stroup, 2011), permeability estimated from poroelastic triggering of earthquakes indicated values from $k \approx 10^{-13}$–10^{-9} m^2, indicative of fracture dominated flow.

Exactly how fluids flow through complex networks is not well known, but the conceptual models of fracture meshes and structural permeability (Hill, 1977; Sibson, 1996; Sibson, 2000) appear the most relevant for forming and maintaining permeable pathways within the crust. Indeed, Enhanced Geothermal Systems (EGS) aim to develop such structures in order to maintain a viable geothermal resource. In the crust, as in EGS systems, injection of high pressure fluids would facilitate shear failure on high strength bridges that are isolated from the (evolving) hydraulic network (Terakawa, Miller, & Deichmann, 2012).

2.2 The Permeability Scale and Toggle-Switch Permeability

The sheer range of the permeability scale (from 10^{-22} to 10^{-8} m^2) makes it very difficult to constrain, particularly considering that changes of that order are possible for common tectonic processes. For example, prior to an earthquake, trapped, high pressure fluid is entombed in a system of low permeability that restricts diffusion and flow. When the earthquake occurs, diffusion and flow are described by Eq. (1.3). Where serpentine or other rocks with positive Clapeyron slopes dehydrate, or in melting reactions (Connolly, 1997), the reaction itself produces more fluid than the created porosity can store, promoting local tensile hydro-fracture at the nucleation site (Miller et al., 2003).

The inherent complexity of permeability can be simplified to an on—off "toggle switch" permeability (Miller & Nur, 2000), which results in a self-organizing system with a power law relationship of connectivity at the percolation threshold. In this simple model, pockets of fluids interact only with their nearest neighbors, and connectivity evolves in response to some stimulus (e.g. earthquake, hydrofracture, etc.), producing a rich complexity inherent in all self-organizing models (Bak, Tang, & Wiesenfeld, 1987). Figure 1.1 shows model results of the connectivity of fluid pressure pockets at the percolation threshold, and helps to visualize possible fluid pressure distributions within fault zones or within dehydrating systems. Figure 1.1 also illustrates how fluids provide an additional natural mechanism for inherent strength heterogeneity due to variations in the effective normal stress acting on the interface. Since such cellular automata have no length scale, this visualization could represent states of fluid pressure from the pore scale to the crustal scale.

2.3 Fluid Flow in Ductile Environments

The most efficient, and perhaps dominant, process for mobilizing fluids in ductile environments is via porosity waves (Connolly, 1997; Connolly & Podladchikov, 2004, 2007). Porosity waves, first discussed as solitary waves (Richardson, Lister, & McKenzie, 1996; Scott, Stevenson, & Whitehead, 1986), are packets of elevated interconnected and fluid-filled porosity that travel as shape-preserving (shock) waves within a ductile matrix. This buoyant packet propagates in the direction of prevailing pressure gradients at a velocity controlled by the viscosity of the entraining matrix and the compaction length. Figure 1.2 shows a numerical example of a porosity wave, demonstrating the sharp front and elongated tail of fluid-filled porosity.

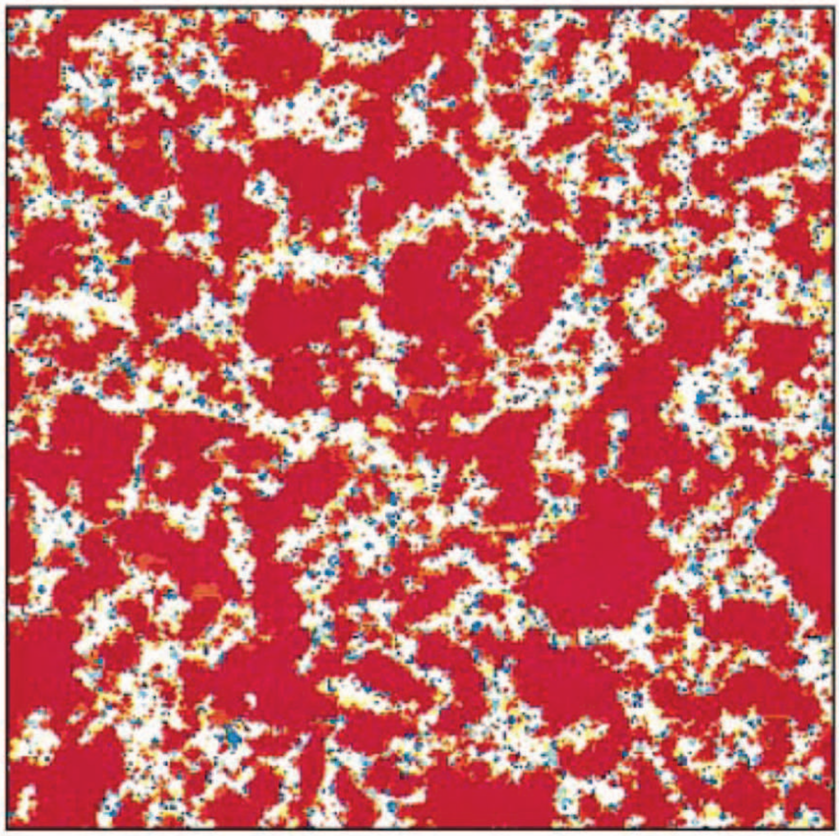


Figure 1.1 Connected compartments at percolation threshold where permeability is treated as a toggle switch in a cellular automaton model (Miller & Nur, 2000). Red (or darker) designates regions where fluid pressure is at or near a critical threshold (e.g. slip, or hydrofracture, etc.). Reprinted with permission from Elsevier. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this book.)

This mechanism of fluid transport, developed for application to compacting sedimentary basins (Connolly, 1997), has been extended to the lower crust to show how long-lived pockets of fluid pressure can be maintained (Connolly & Podladchikov, 2004), and how negative pressure gradients below the brittle-ductile transition can drive fluids into the lower crust. This is potentially important for the later discussion of fluid-driven aftershocks

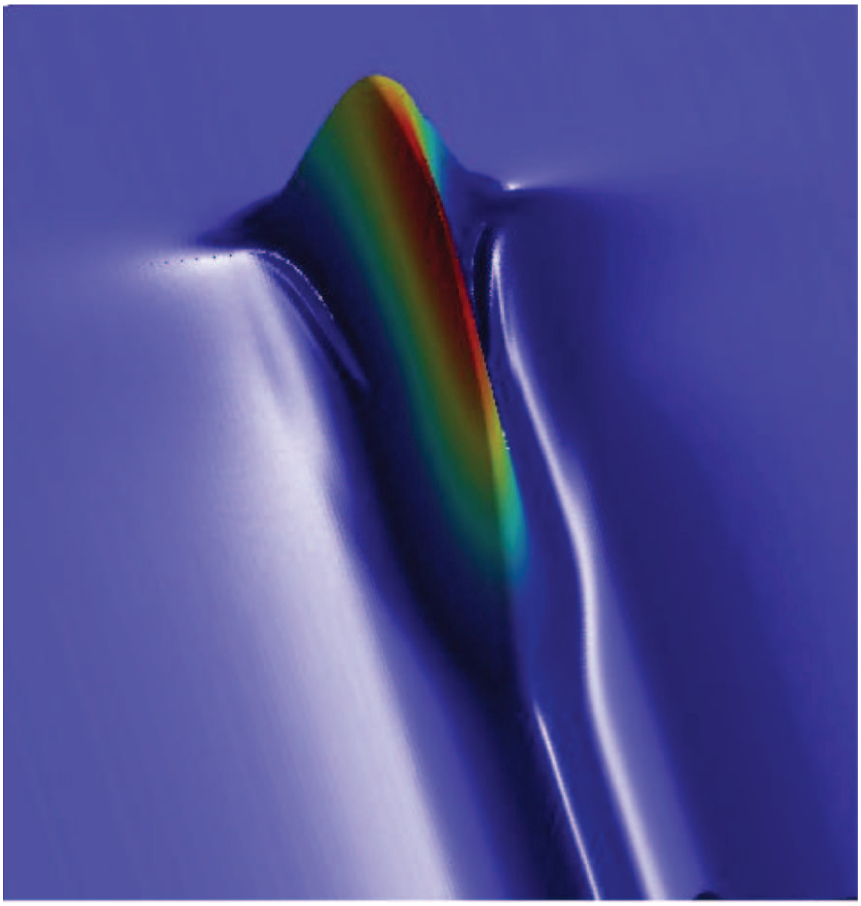


Figure 1.2 Two-dimensional numerical simulation (from Connolly and Podladchikov (2007)) of a quasi-steady state porosity wave propagating through a viscous matrix that decompacts more easily than it compacts. The wave propagates in the direction of the hydraulic gradient (the mean stress gradient less the hydrostatic gradient of the pore fluid) and gains fluid mass from the matrix with time as a consequence of the asymmetric pressure distribution that arises in decompaction-weakening rheology. The wave depicted here has just initiated and is roughly two compaction lengths long and has a maximum amplitude that is twice the background porosity. Wavelengths can grow to 10–15 compaction lengths, while amplitudes can reach 100–1000 times background porosity. Figure courtesy of J.A.D. Connolly. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this book.)

because it provides a means of establishing neutrally buoyant pods of over-pressured fluids that can remain stable over long time periods until breached co-seismically from a deeply penetrating fault system.

3. FOLLOW THE FLUIDS

To follow the fluids is to follow the tectonic cycle. Figure 1.3 schematically illustrates where fluid-rock interactions are most active, and serves as a roadmap for the paper. The most abundant fluids affecting tectonic processes are H_2O and CO_2, trapped as hydrous minerals in metabasalts, sediments, serpentinites, and carbonates. There are numerous chemical reactions to consider, but a few important ones are given here to illustrate common reactions that either consume or release fluids (Kelemen et al., 2011).

Consumption of H_2O through the reaction

$$\begin{aligned} &2\mathrm{Mg_2SiO_4} + \mathrm{Mg_2Si_2O_6} + 4\mathrm{H_2O} \rightarrow 2\mathrm{Mg_2Si_2O5(OH)_4}, \quad (1.4)\\ &\text{Mg-Olivine} + \text{Mg-Pyroxene} + \mathrm{H_2O} \rightarrow \text{Serpentine} \end{aligned}$$

occurs wherever water encounters mantle peridotite, such as in mid-ocean ridges, along fracture zones, or where free fluids encounter the mantle wedge.

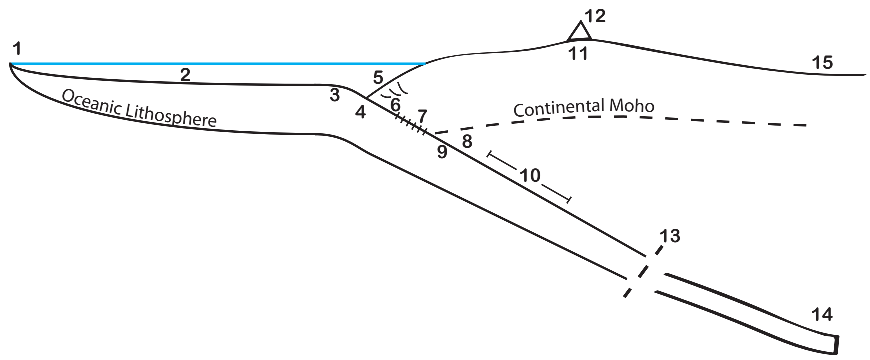


Figure 1.3 Simplified schematic (not to scale) of fluid involvement in tectonic processes. (1) Hydrothermal circulation (serpentinization) at MOR, (2) Carbonate sedimentation, entrainment of pore fluids, (3) Bending of outer rise (normal faulting and hydrothermal alteration), (4) Thermal profile entering subduction, (5) Terrigenous sediment input and accretionary prisms, (6) Dehydration of clays and low pressure metamorphism, (7) Locked interface, (8) Serpentinization of the mantle wedge, (9) Episodic tremor and slow slip earthquakes, (10) Dehydration of serpentinite and hydrous partial melting of mantle wedge, (11) Large strike-slip faults hosting (12) Volcanic arcs, (13) Slab breakoff, (14) Melting of pelites and CO_2 production, and (15) Intraplate earthquakes. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this book.)

Water becomes a free fluid when liberated by the dehydration reaction:

$$\begin{aligned}&2Mg_3SiO_5(OH)_4 \rightarrow 3Mg_2SiO4 + SiO_2 + 4H_2O,\\&\text{Serpentine} \rightarrow \text{Mg-Olivine} + \text{Silica} + H_2O,\end{aligned} \tag{1.5}$$

which occurs in the subducting slab at appropriate *P-T* conditions that depend on many factors including slab age, convergence rate, and frictional heating.

CO_2 can be sequestered via:

$$\begin{aligned}&2Mg_2SiO_4 + 2CO_2 \rightarrow 2MgCO_3 + SiO_2,\\&\text{Mg-Olivine} + CO_2 \rightarrow \text{Magnesite} + \text{Quartz}\end{aligned} \tag{1.6}$$

and is important in the hydrothermal circulation systems ubiquitous at mid-ocean ridges (MOR), where in the presence of water, CO_2 is sequestered via:

$$\begin{aligned}&Mg_2SiO_4 + CaMgSi_2O_6 + 2CO_2 + 2H_2O\\&\quad \rightarrow Mg_2Si_2O5(OH)_4 + CaCO_3 + MgCO_3,\\&\text{Mg-Olivine} + \text{Ca-Mg-Pyroxene} + CO_2 + H_2O\\&\quad \rightarrow \text{Serpentine} + \text{Calcite} + \text{Magnesite}.\end{aligned} \tag{1.7}$$

CO_2 is released through the decarbonization reaction at 5 GPa (Hammouda, 2003):

$$\begin{aligned}&4CaCO_3 + 4SiO_2 + (Mg,Fe)_3\,Al_2Si_3O_12\\&\quad \rightarrow 3Ca(Mg,Fe)Si_2O6 + CaAl_2SiO_6 + 4CO_2,\\&\text{4 calcite} + \text{4 coesite} + \text{garnet}\\&\quad \rightarrow \text{3 clinopyroxene} + \text{Tschermack's molecule} + 4\ CO_2\end{aligned} \tag{1.8}$$

while in the presence of water decarbonation follows:

$$\begin{aligned}&CaCO_3 + SiO_2 \rightarrow CaSiO_3 + CO_2,\\&\text{calcite} + \text{quartz} \rightarrow \text{wollastonite} + \text{carbon dioxide}.\end{aligned} \tag{1.9}$$

All of these reactions depend on pressure and temperature, and will occur at different places and times during the tectonic cycle. As temperature-dependent processes, they are linked to the cooling oceanic lithosphere because the age of the lithosphere determines the temperature profile ultimately subducted. Much literature focusses on the reactions in subducting plates, and interested readers should consult (Hyndman & Peacock, 2003; Kerrick, 2001; Kerrick & Connolly, 2001a; Kirby, 2000; Peacock, Christensen, Bostock, & Audet, 2011; Rupke, Morgan, Hort, & Connolly, 2004). The essence of these studies is that, depending on the thermal structure of the oceanic lithosphere, volatiles are released continuously (i.e. not focused at particular isograds) during subduction, with significant mechanical and chemical consequences.

3.1 Fluids and Mid-Ocean Ridges

New lithosphere created at mid-ocean ridges (MOR) is highly altered by hydrothermal circulation of ocean water through fractured rock. Reaction of sea water with the boundaries of the fractured system creates serpentinized peridotite that can store large quantities of H_2O. These complex interacting processes of heat flow, fracturing, fluid flow, and porosity creation/destruction are conceptually well-understood, but attempts at constraining this dynamical system are hampered by significant and transient changes in the underlying physical parameters needed to quantify the system. Intense hydrothermal circulation affects earthquake genesis and statistical properties (Bohnenstiehl, Waldhauser, & Tolstoy, 2008), while intense fracturing from thermal and tectonic stresses measured near the East Pacific Rise indicate highly permeable pathways for fluid rock interactions (Tolstoy, Waldhauser, Bohnenstiehl, Weekly, & Kim, 2008). At the MOR, the reactions described in Eqs. (1.4) and (1.6) occur, sequestering H_2O and CO_2 in large quantities, with meta-basalts containing almost 3 wt.% H_2O and 3 wt.% CO_2 (Kerrick, 2001). The process reverses to produce H_2O and CO_2 at a time and place determined by the thermal structure of the awaiting subduction zone. Figure 1.4 (Connolly, 2010) shows an example *P-T* diagram for meta-pelite, with different *P-T* paths for fluid release depending on the angle of subduction (*P*), and the thermal structure (*T*). Similar figures can be drawn for many rock types, changing only in the *P-T* details. The depth and degree of alteration depends on the spreading velocity, with significantly deeper and extensive alteration in slow spreading centers because of the ability for deep penetration of water through fractures created in normal faults of the colder lithosphere (Poli & Schmidt, 2002). Although not well constrained, estimates of the depth of pervasive alteration are on the order of 500 m (Kerrick, 2001), while significantly deeper penetration might be expected for larger fault systems. The extent of sea water percolation into thermally fractured lithosphere is controlled by the permeability of the fracture network, which increases significantly when the fracture or

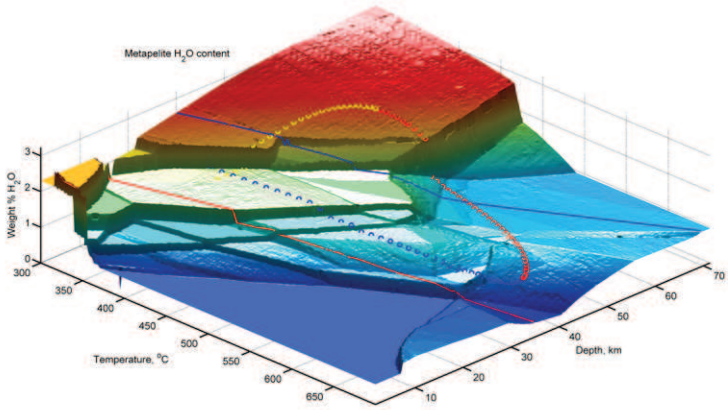


Figure 1.4 Water-content for average pelitic sediment as a function of temperature and pressure computed assuming equilibrium with a pure H_2O fluid. Red and blue lines, respectively, indicate hot (20 K/km) and cold (10 K/km) metamorphic geotherms. From (Connolly, 2010), with permission from the Mineralogical Society of America. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this book.)

fault forms, but which then shuts down when the system becomes clogged by the reaction itself (Jamtveit, 2010). Permeability measurements of oceanic basalt range from 10^{-13} to 10^{-9} m^2 (Fisher, 1998), indicative of highly fractured systems. Such high permeabilities allow pervasive flow and fluid-rock interactions, with volatiles held in long-term storage until eventual release within the subduction zone.

3.2 Sedimentation on the Ocean Floor

As mantle convection drives oceanic lithosphere towards its subduction demise, the volatile package created at the MOR is covered in pelagic carbonate sediments and pore- and chemically-bound H_2O. Carbonate is abundant in two main pelagic marine sediments: (1) siliceous limestones and (2) clay-carbonates (marls) (Kerrick & Connolly, 2001a). The additional pelagic sediments and terrigenous sediments transported from continental margins via turbidity currents contain large quantities of CO_2 and H_2O, but the thickness of these deposits is variable and generally not well-constrained. The depth of sediment deposits varies along subduction zones, where sediments are mostly derived from the continents, while sedimentation in deep oceanic basins is poorly known because of limited sampling from ocean drilling.

3.3 Fluids and the Outer Rise

Flexural bending of the oceanic lithosphere produces a bulge oceanward from the trench known as the outer rise. As in any bending problem, extensional stresses develop at the top of the plate and compressional stresses at the base of the plate. How much it deflects, and therefore how much it bulges at the outer rise, depends on the flexural rigidity D, a material property predominantly controlled by the plate thickness, h, where $D \propto h^3$ (Burov & Diament, 1995). Old and cold lithosphere, being much thicker, produces less bending than thinner oceanic lithosphere associated with newer and warmer plates. In either case, bending facilitates normal faulting near the outer rise, and the depth to which faulting occurs determines the depth to which serpentinization occurs, just as near the MOR. Seismic evidence for plate-bending induced faulting in Figure 1.5 indicates pervasive serpentinization of the oceanic crust and into the mantle. Later dehydration of this plate in the subduction zone has been suggested as a mechanism for intermediate depth earthquakes (Ranero, Morgan, McIntosh, & Reichert, 2003).

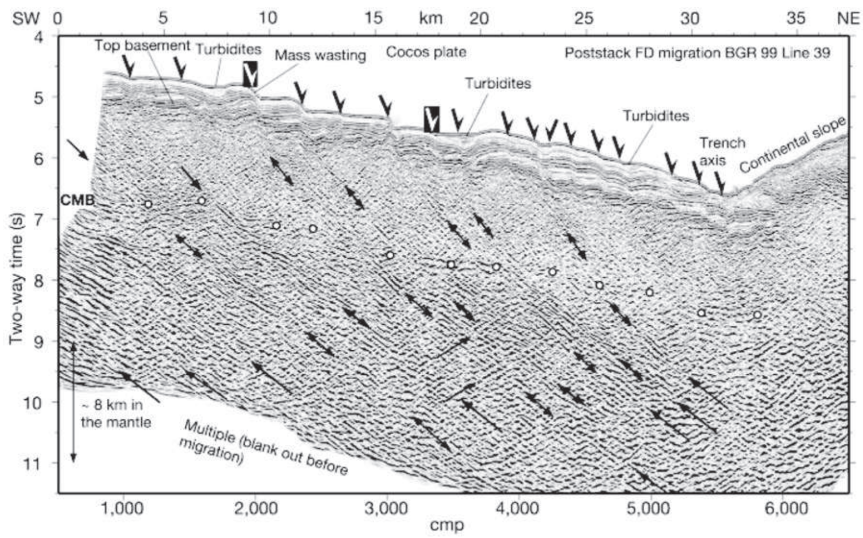


Figure 1.5 (a) Seismic profiles across the Cocos plate in Costa Rica showing evidence of normal faults due to plate bending. Such faulting would allow serpentinization for future dehydration and triggering of intermediate depth earthquakes. Reprinted with permission from Macmillan Publishers Ltd: Nature (Ranero et al., 2003).

Evidence suggests that stress release at the outer rise in response to nearby mega-thrust subduction zone earthquakes can initiate large-scale normal faulting in the rise (Ammon, Kanamori, & Lay, 2008; Beavan et al., 2010). Interestingly, this produces a situation where subduction zones essentially feed themselves with future water by triggering outer-rise earthquakes that gives rise to large volumes of additional fluid through serpentinization (Kirby, Stein, Okal, & Rubie, 1996).

There has been significant debate about the depth extent of this outer-rise alteration, because the depth of alteration affects whether devolatilization is potentially responsible for both upper and lower plate seismicity of the subducting slab (Peacock, 2001). Double subduction zone seismicity long remained enigmatic because it was unclear how serpentinization could extend to the deeper regions of the oceanic lithosphere. An intriguing explanation, which also fits with the notion of almost exclusively dehydration-assisted seismicity in subduction zones, comes from numerical modeling (Faccenda, Gerya, & Burlini, 2009). Figure 1.6 shows that tectonic stresses associated with lithospheric bending can allow fluids to penetrate deeper into the oceanic lithosphere. Although fluids are always buoyant, fluid flow responds to pressure gradients, so in regions where the pressure gradient is in the direction of gravity, fluids will be driven to deeper levels. This modeling has been further developed (Faccenda, Gerya, Mancktelow, & Moresi, 2012), showing the same effect is found within subduction zones during unbending, supporting dehydration as viable mechanism to explain double-seismic zone seismicity. This same mechanism of tectonic pressures driving fluid flow in the direction of gravity was found for fluids trapped in the lower crust (Connolly & Podladchikov, 2004). Namely, near the brittle-ductile transition (BDT), pressure gradients can act in the direction of gravity to create isolated pockets of over-pressured fluids within the lower crust.

3.4 Thermal Profile Entering Subduction

Having collected, entrained and stored varying volumes of fluids, the subduction package begins the descent that eventually releases the fluids. The speed of the reversal depends mostly on the convergence rate and thermal structure, which is controlled by age. Two popular end-member cases are the young, hot subduction zones of Cascadia and southwest Japan, and the older, cold subduction of northeast Japan beneath Honshu Island and the site of the 2011 Tohoku earthquake (Keken, Hacker, Syracuse, & Abers, 2011; Peacock & Wang, 1999; Syracuse, Keken, & Abers, 2010). Figure 1.7 shows the two Japanese steady state temperature profiles, which depend on boundary

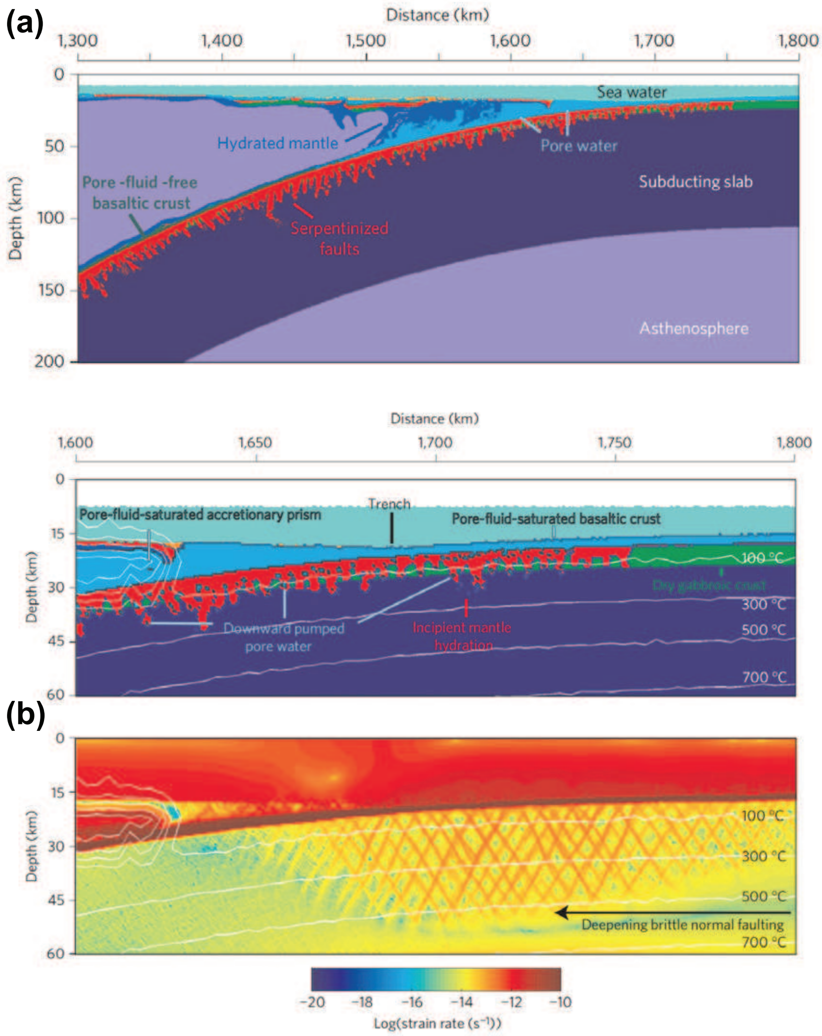


Figure 1.6 (a) Top panel shows serpentinized faults being subducted, with a zoomed view in the lower panel. (b) Deepening of normal faulting due to bending at the outer rise, resulting in pumping of fluid deeper into the oceanic mantle. Reprinted with permission from Macmillan Publishers Ltd: Nature Geoscience (Faccenda et al., 2009). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this book.)

condition assumptions near the mantle wedge and down-dip from the thrust zone. All models suggest that, except for the coldest of slabs, complete dehydration of H_2O occurs before the plate arrives to sub-arc depths. The dip

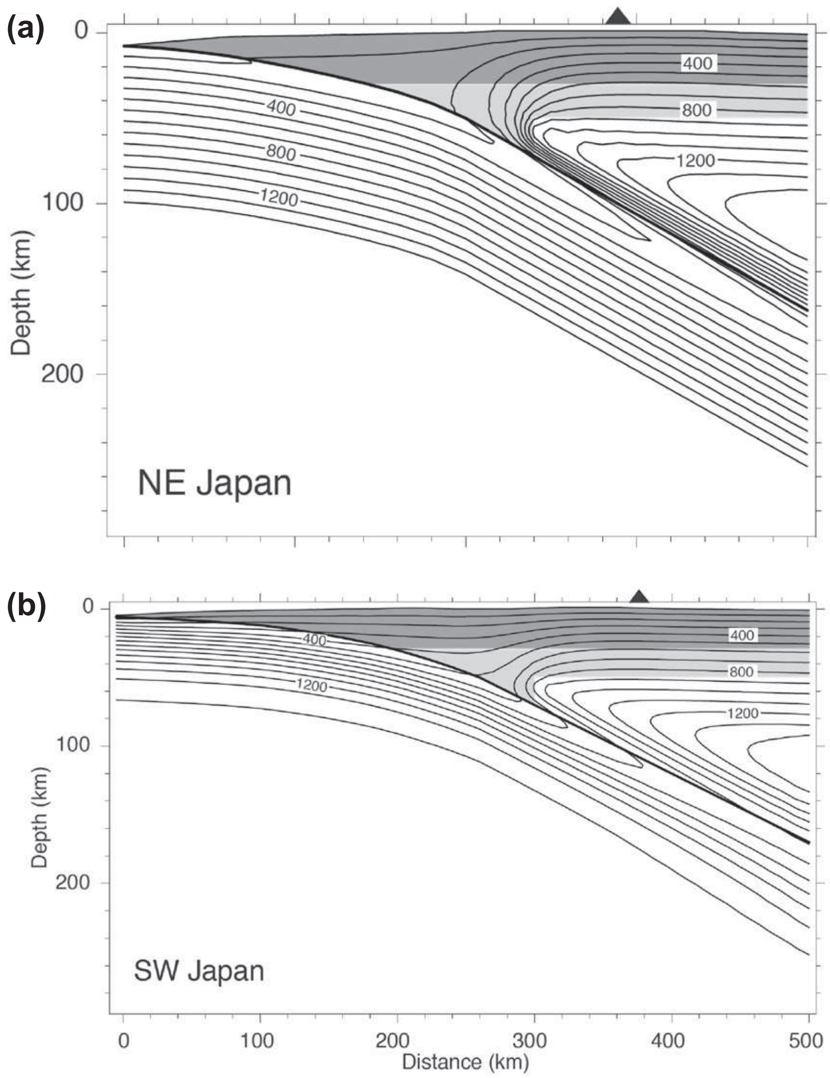


Figure 1.7 Thermal profiling being subducted in (a) Northeast and (b) Southeast Japan. Reprinted from Peacock and Wang (1999) with permission from AAAS.

angle of the subducting thermal profile controls the depth at which dehydration occurs, with the hot and young Cascadia subduction zone (not shown) completely dehydrating at depths of around 100 km, while the old and cold Pacific plate largely avoids substantial dehydration to those depths.

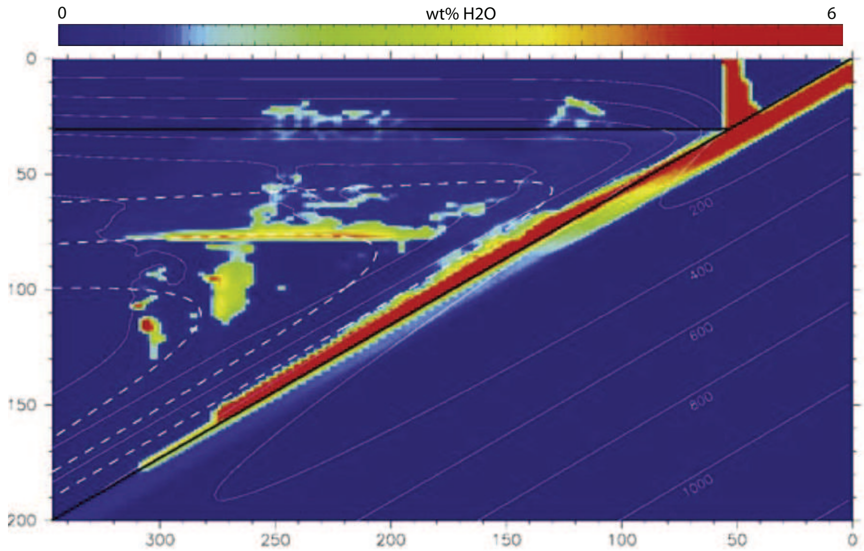


Figure 1.8 Numerical model predictions of H_2O within the subduction system of NE Japan, From (Iwamori, 2007) with permission from Elsevier. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this book.)

In northeast Japan, which hosts the cold subduction, very little dehydration is expected even though large quantities of H_2O are present within the subducting package. Modeling and seismic velocity measurements in NE Japan show significant transport of H_2O to the deep mantle within a fully hydrated oceanic crust and a serpentinization of the mantle wedge (Iwamori, 2007). Figure 1.8 shows where hydrated minerals are expected and, because the plate is cold, little dehydration is expected. One can speculate that the lack of aqueous fluid available to weaken the plate interface may have contributed to the large amount of slip released in the Tohoku earthquake, since the degree of locking should be directly related to the availability of fluids. That is, with no free fluids available to weaken the interface, much higher slip deficits may accumulate. Claims that the Pacific plate avoids dehydration, however, seem to conflict with tomographic studies performed using the Tohoku aftershocks, which show a substantial influence of fluids (presumably from dehydration) on that earthquake sequence (Wang, Huang, Zhao, & Pei, 2012).

The estimated temperature profiles entering subduction rely on the heat transport assumptions used in the numerical modeling. There are differing degrees of sophistication in the modeling, with pure conduction the easiest and simplest approximation. Models with greater complexity include frictional heating, radioactive decay, mantle wedge corner flow, and the potentially important element of heat advection through fluid flow within fracture networks (Spinelli & Wang, 2009). Models that include heat advection via fluid flow show that the down-going plate can be significantly cooled, resulting in important dehydration phase boundaries being reached at substantially deeper levels in the subduction zone. Advection models that match observed heat flow (Spinelli & Wang, 2009) call for very high permeability in the oceanic crust, consistent with measured permeability of oceanic basalt of 10^{-9}–10^{-10} m^2, corresponding to average crack apertures of around 10^{-4} m when using Eq. (1.3).

Fluids that ultimately reach the subduction zone interface face one of three fates: (1) Pore fluids are squeezed out of the pore space and invade into the accretionary prism; (2) hydrous minerals are subducted and released by dehydration at progressively deeper levels of the subducting package; or (3) fluids are brought deeper into the mantle. These fates depend on many factors, including the thickness of sediments arriving at the convergent boundary, the degree of hydrothermal alteration within the oceanic crust, and the temperature profile of the subducting slab. The weakening effect of fluid is important not only for subduction dynamics; indeed, modeling results show that subduction initiation itself is difficult to achieve without a substantial weakening mechanism such as the presence of water (Gerya, 2011; Regenauer-Lieb, Yuen, & Branlund, 2001).

3.5 Into the Trench

The thickness of the sediment package entering subduction zones varies among subduction zones and even along portions of the same subduction zone. For example, the thickness of sediments entering the southern Chilean subduction zone extends to upwards of 1000 m, while in northern Chile sediments are virtually absent. Presumably, this variability is related to climatic-driven erosion of the continental margin, which also impacts the evolution of accretionary prisms (Simpson, 2010). Other sediment rich subduction environments include Cascadia, the Nankai Trench, Northern Sumatra, and Barbados. Each of these regions has a well-developed accretionary prism, while sediment-poor regions like northern Chile and Tonga (Poli & Schmidt, 2002) do not host well-developed accretionary prisms. The sediment supply should influence subduction style (Vonhuene & Scholl, 1991), and possibly their respective arcs, because of the large volumes of pore-trapped fluids. The subducted sediment package and the down-going thermal profile also influence shallow thrust earthquakes in subduction zones because of dehydration of smectite and illite between 100 and 200 °C. Low temperature reactions of clays, zeolites and chlorites are usually finished within the first few kilometers of the subduction zone (Moore & Vrolijk, 1992).

3.6 Accretionary Prisms

The interface between the accretionary wedge and the subducting oceanic lithosphere is weak, although this region is capable of generating some moderate sized earthquakes (Moore & Vrolijk, 1992; Saffer & Tobin, 2011) as well as "slow, tsunami earthquakes" (Kanamori & Kikuchi, 1993). The interface between the accretionary prism and the subducting plate is dominated by clay-rich sediments, which were long thought to be velocity strengthening and not capable of sustained instability. However, recent experimental results suggest that, although the clay-rich sediments are predominantly and strongly velocity strengthening, at high slip velocity these same sediments show velocity weakening behavior and thus capable of sustaining unstable earthquake ruptures (Faulkner, Mitchell, Behnsen, Hirose, & Shimamoto, 2011).

Focussed flow is common in accretionary prisms (Carson & Screaton, 1998), and faulting allows the escape of trapped fluids that drain the interface of elevated fluid pressure. A natural consequence of this mechanism is that the frictional properties of the accretionary wedge interface are highly heterogeneous, which can be manifested as complex slip distributions along accretionary prism interfaces. Conceptually, Figure 1.1 provides a visualization of how heterogeneous slip distributions may result if high pressure fluid leaks into the third dimension of the accretionary prism, leaving behind yet un-tapped pockets of overpressured fluids (and thus weak zones) along the thrust interface. Observations from Cascadia show that faults and/or diapiric structures have brought hydrocarbons from more than 1.5 km depth to the surface in the last 400 years (Carson & Screaton, 1998), indicative of fast fluid migration. Direct observations of the Oregon accretionary prism show that most fluid flow is focussed along strike-slip faults, suggesting that draining of the overpressured fluid increases the strength of the interface, resulting in thrust faulting (Tobin, Moore, Mackay, Orange, & Kulm, 1993).

Strike-slip faulting provides an efficient means of draining over-pressures developed along the margin, thus strengthening regions that previously slipped freely but are now capable of locking. Indeed, splay faults in the accretionary prism have been proposed as fluid migration pathways (related to aftershock generation) following the 2004 Andaman Earthquake in Sumatra (Waldhauser, Schaff, Diehl, & Engdahl, 2012), with yet another example of very high inferred permeability of 10^{-13}–10^{-10} m^2.

3.7 The Locked Zone

The locked zone (see Figure 1.3) is responsible for mega-thrust subduction zone earthquakes capable of wide-spread destruction. Heterogeneous strength is inferred within the locked zone (Moreno, 2012; Moreno, Rosenau, & Oncken, 2010) and is attributed to variations in frictional strength. One possible explanation for the large along-strike variation in strength is that fluids participating in a connected hydraulic network represent weak zones, while regions disconnected from the hydraulic network provide for the interface strength and are known as asperities.

Numerical modeling of shear stresses on subducting plate interfaces show that a shear resistance of at least 20 MPa is needed to support the overlying topography (Lamb, 2006). The actual shear stress acting on different subduction zones depends on the geometry, the convergence rate, and the availability of sediments to subduct. These factors affect the degree of coupling between subduction zones and even within individual subduction zones, while other processes such as trench rollback have also been suggested to influence coupling (Scholz & Campos, 1995). Interestingly, sediment-starved margins have substantially higher shear stresses and support much higher topography than their sediment-rich counterparts (e.g. southern Chile versus northern Chile), further supporting the notion that fluid availability is the dominant property controlling subduction style.

3.8 The Mantle Wedge

As subduction proceeds, cold oceanic lithosphere encounters the hot mantle wedge. Free fluids brought to this interface are quickly ingested, resulting in a serpentinized mantle wedge. This important interface determines the down-dip extension of locked zones because it constrains the down-dip rupture dimension, and thus potential earthquake magnitude. The serpentinized mantle wedge is probably very weak. One form of serpentine, antigorite, has been tested in the laboratory under the conditions expected at the mantle wedge (Hilairet et al., 2007). These experiments showed that the stress exponent that relates stress to strain rate is on the order of six, resulting in very high creep rates at low shear stresses. The serpentinized mantle wedge also acts as an impermeable barrier to flow, allowing fluids created through dehydration to pool along the interface. Episodic tremor and slow slip earthquakes occur near the mantle wedge, making this junction is a very important component in the subduction process that will undoubtedly receive further attention.

3.9 Episodic Tremor and Slow Slip

Very good and extensive reviews of episodic tremor and slip (ETS) are already in the literature (Beroza & Ide, 2011; Gomberg, 2010; Peng & Gomberg, 2010), so an in-depth review here is unnecessary. Relevant here is that the overarching conclusion from the last decade of ETS observations is that high pressure fluids are involved. This conclusion is mostly based on observed elevated Vp/Vs ratios (Shelly et al., 2007) and that ETS is triggered by very small changes in stress associated with passing surface waves from distant earthquakes (Rubinstein et al., 2009). Additionally, non-volcanic tremor has the same characteristics as volcanic tremor, which is widely recognized as associated with the movement of fluids. The current physical models being investigated include rate and state friction (Liu, Rice, & Larson, 2007; Liu & Rubin, 2010; Segall, Rubin, Bradley, & Rice, 2010; Shibazaki, 2005; Shibazaki & Iio, 2003) and a stochastic process of cascading failures of asperities failing in a random walk (Ide, 2008). Figure 1.9 is the current paradigm for trying to understand how fluids are involved in this important phenomenon (Shelly, Beroza, Ide, & Nakamula, 2006). Future research may require additional mechanisms including dehydration (Brantut, Sulem, & Schubnel, 2011), and more complicated rheological models because ETS appears to occur near the brittle-ductile transition.

3.10 Feeding the Arc

A long-held view called for a direct causal relationship between dehydration and melting of the mantle wedge, primarily via dehydration of chlorites in the mantle wedge and subducting lithosphere (Grove, Till, Lev, Chatterjee,

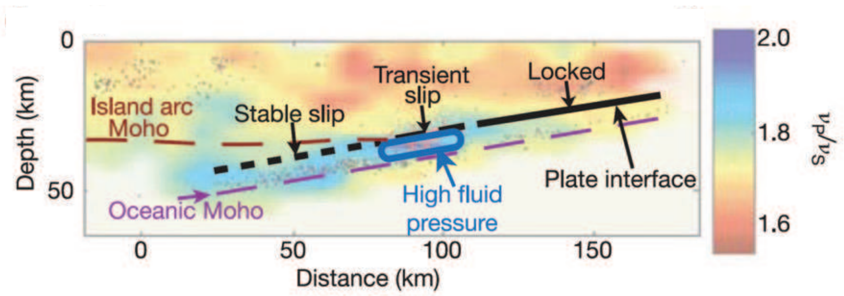


Figure 1.9 Conceptual model of trapped over-pressured fluids as an integral part of the processes controlling ETS. Reprinted with permission from Macmillan Publishers Ltd: Nature from (Shelly et al., 2007). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this book.)

& Medard, 2009). This view has recently been questioned (England & Katz, 2010), and the argument is made for a fully thermally-controlled system whereby only the hottest portion of the mantle wedge contributes to arc volcanism. The key observation to support the dehydration model is the sharp boundary of volcanic arcs and a relatively tight range of trench-arc distances. However, a suite of models (Syracuse et al., 2010) showed that in all but the coldest of subduction zones, all of the sediments and most of the top of the oceanic crust would have dehydrated before reaching depths below volcanic arcs. The apparent absence of dehydration makes it difficult to reconcile the amount of CO_2 exhaled during arc volcanism (Kerrick et al., 2001b; Zhang & Cox, 2000), and how this occurs is still an open question.

3.11 Strike-Slip Faults and Volcanic Arcs

In both southern Chile and Sumatra, the volcanic arc resides on tectonic-scale strike-slip faults, while through-going faults along other arcs are not as obvious. The coexistence of strike-slip faults and extensive volcanism is probably not a coincidence. A numerical study linking transform faulting and the development of piercement structures, related to the LUSI mud volcano in Indonesia, shows that shear induced reductions in the effective viscosity can mobilize buoyant fluids and produce piercement structures similar to what is observed in southern Chile and Sumatra. Vertical fault planes may in general host an abundance of deeply derived fluids because they "provide a short-circuit to the mantle" (Greg Hirth, Brown University, personal communication, 2011).

3.12 Slab Break Off, Carbonate Melting, and Intraplate Earthquakes

The dynamics of slab break off are relatively well-known through numerical experiments (Duretz, Gerya, & May, 2011; Gerya, Yuen, & Maresch, 2004), and the consequence of this break off is a rapid drop of the plate into the mantle. The sinking slabs will continue to be heated and pressurized and therefore degassed (Gorman, Kerrick, & Connolly, 2006). Melting of carbonates is difficult in the absence of water (Grassi & Schmidt, 2011), so large volumes of carbonate are brought to very deep depths. The fate of the CO_2, once produced, is not known, but it either stays stored in the mantle or returns to the surface. A catalog of subducted plates (van der Meer, Spakman, Hinsbergen, Amaru, & Torsvik, 2010) shows remnants of subduction since the Permian, including the Farallon Plate beneath North America, while slab break off in the Mediterranean-Carpathian region (Wortel & Spakman,

2000) is imaged below the Eurasian plate. It is intriguing to postulate that detached, descending, and decarbonating subducting plates contribute to intra-plate seismicity with intra-plate faults providing degassing pathways for the high pressure CO_2 produced at depth. Specifically, carbonates in the absence of water are very difficult to melt until deep in the mantle, so when they eventually do melt, the CO_2 would migrate towards the surface over very long time periods. When they reach the surface, they are far from the original plate boundary and could very likely reside beneath continental lithosphere like evidenced from the Farallon plate.

4. FLUIDS IN THE EARTHQUAKE PROCESS

It is clear from the previous discussion that substantial fluid is released at depth with minimal hydraulic connectivity to the free surface. As such, the pressure of the fluid is close to the lithostatic load. Thus the influence of fluids on lithospheric dynamics is significant, and fluids may be a primary driver of earthquake cycles.

4.1 Fluids along Frictional Interfaces

Early and prescient works (Hubbert & Rubey, 1959; Mase & Smith, 1984, 1987; Nur & Booker, 1972; Sibson, 1973) recognized that trapped fluids within fault zones would have important effects on earthquakes, either by reducing the effective normal stress through high pore pressure, or by generating high pore pressure from frictional heating of the fault zone. Whether high pore pressures are generated and maintained, or dissipate quickly, depends on local permeability within the fault and damage zone (Mitchell & Faulkner, 2009; Faulkner et al., 2011). Lateral variations in pore pressure within the fault zone (Byerlee, 1993) could result in a complex topology of isolated pockets of trapped high pore pressure, and a highly heterogenous distribution of effective normal stress. A numerical model of this scenario (Miller, Nur, & Olgaard, 1996) did indeed show a rich heterogeneity in fault properties. Fault zones hosting deeply derived CO_2 are evidenced by rich travertine deposits found in many fault zones in Turkey (Uysal et al., 2009) and Italy (Chiodini et al., 2004). Evidence for fluids in fault zones, although ubiquitous, is still difficult to interpret because of the myriad of processes acting, including dissolution/precipitation (Gratier et al., 2011), hydrofracture and slip-related changes in hydraulic properties, dehydration, poro-elastic stresses, and complicated feedbacks with the thermodynamics of a slipping frictional interface.

4.2 Rate-State Friction

The sliding of frictional interfaces is well-described by the rate- and state-dependent friction constitutive model (RS-friction), with a vast literature spanning the decades since this behavior was first observed in laboratory experiments (Dieterich, 1979; Dieterich, 1992). Developed for dry interfaces, the RS form of friction is

$$\tau = \mu\bar{\sigma}_n = \left(\mu_o + a\ln\frac{V}{V_o} + b\ln\frac{V_o\theta}{D_c}\right)\bar{\sigma}_n, \tag{1.10}$$

where a and b are laboratory-determined constants, with $a - b > 0$ called velocity strengthening (stable slip) and $a - b < 0$ called velocity weakening (unstable slip); V is the step change in sliding velocity from an initial velocity V_o; D_c is the critical slip distance; $\bar{\sigma}_n$ is the effective normal stress (normal stress minus fluid pressure); μ_0 is the reference friction, and θ is the state variable that evolves as

$$\frac{d\theta}{dt} = 1 - \left(\frac{V\theta}{D_c}\right). \tag{1.11}$$

Although this relationship cannot be derived from first principles, the heuristic arguments for the state variable as a measure of memory of asperity topology are intuitive (Scholz, 1998). In this argument, the state variable evolves over a critical slip distance to its steady state sliding velocity because it needs a certain distance to forget the initial roughness and population of contacts of the interface from where it started to slip. This formulation well-describes the sliding behavior of a very wide range of materials. An important aspect of Eq. (1.10) is that friction acts on the effective normal stress. At elevated pore pressures, the stability of the interface is still controlled by $a - b$, but when fluid pressures approach some percentage of the normal stress, then the frictional interface slides stably, disallowing earthquake instability (Segall & Rice, 1995). Another important control is the degree of dilatancy as the rough asperity rides over the oncoming topology, with large dilatancy allowing fluid pressure reductions and thus leading to instability. The conditionally stable state, which depends on the dilatancy, can either become stable sliding or unstable sliding depending on the complex interactions of the (poorly-constrained) dilatancy coefficient (Hillers & Miller, 2006, 2007; Segall & Rice, 1995).

The power of RS friction, and its Achilles heel, is that it provides much latitude in possible behaviors, and can therefore be used to describe the dynamics of many complex systems, including earthquake nucleation (Dieterich, 1979), afterslip (Perfettini & Ampuero, 2008), and slow-slip earthquakes

(Liu & Rice, 2005; Liu et al., 2007; Segall et al., 2010; Shibazaki, Garatani, & Okuda, 2007). However, convincing manifestations of RS friction in observed earthquake behavior remain elusive. Observations of slow slip preceding the 2011 $M = 9$ Tohoku Japan earthquake (Kato et al., 2012) showed some aspects of RS friction, in that there was some evidence for impending instability, or instabilities, leading to dynamic rupture, but the predicted accelerating (power-law) slip prior to instability predicted by RS friction was not observed. Interestingly, in a study of the impending Tohoku rupture zone prior to that earthquake, seismic evidence showed a hydrated mantle at the subduction zone interface (Kawakatsu & Watada, 2007), which may have been an important contributor to the foreshock-mainshock sequence observed. One possibility is that frictional heating by the foreshock liberated water through dehydration, which drove pressure diffusion within the subduction zone. This is consistent with the observed diffusional character of foreshock migration (Kato et al., 2012).

4.3 Dynamical Influence of Fluids on Earthquakes and Faulting

Slip interfaces during seismic rupture have been shown to be very thin (Sibson, 2003), with most slip is concentrated over only a couple of centimeters in fault core thickness. Therefore, numerical models typically assume a dislocation surface as a reasonable approximation of the slip zone. Numerically generating complexity along those slip zones using RS-friction requires a smattering of heterogeneity, defined in some models as heterogeneous distributions of the critical slip distance D_c (Hillers, Ben-Zion, & Mai, 2006), or through tight constraints on other RS-friction parameters (Shaw & Rice, 2000). Continuous complexity along a planar fault can readily be generated by considering the dynamical interaction between earthquake slip and permeability enhancement co-incident with slip (Miller, 2002; Miller, Ben-Zion, & Burg, 1999; Miller et al., 1996). Step changes in permeability coincident with seismic slip produce a state of continuous complexity along otherwise mathematically smooth dislocation surfaces (Miller, 2002). The complexity emerges, and is sustained, because of the additional degree of freedom in $(\tau, \bar{\sigma}_n)$ stress space associated with co-seismic step changes in local permeability. This additional degree of freedom results in a random walk in stress space, establishing and maintaining a highly disordered stress state that is manifested along the fault plane as stress correlations at all scales, thus generating Gutenberg-Richter (GR) statistics along a single fault plane. Figure 1.10 shows examples of complex slip distributions for spontaneously generated model earthquakes, the magnitudes of which are controlled by

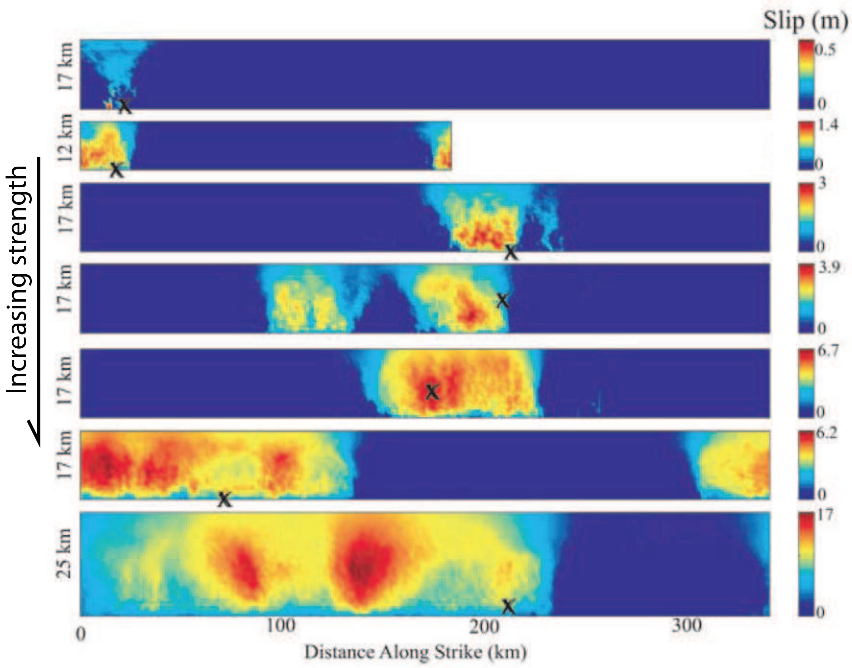


Figure 1.10 Slip distributions for spontaneous ruptures generated in a fluid-controlled fault model. Hypocenters are marked *x*. Fault strength, controlled by the rate of fluid pressure increase, plate velocity, and size of fault increases from the top image (weakest) to the bottom image (strongest). Figure from (Miller, 2002) with permission from American Geophysical Union. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this book.)

the degree of fault overpressure. Real faults are much more complex than those modeled as a dislocation surface, so some caution is necessary when interpreting idealized numerical models.

4.4 Slip at High Velocity

In the fluid-controlled earthquake model described above, fluid sources within the fault plane were assumed to result from inelastic compaction of the pore space (e.g. Walder & Nur, 1984). However, other sources of fluid are available within the fault plane (that is, other processes that can contribute to the $\dot{\Gamma}$ in Eq. (1.2)). These include dehydration reactions of clays and other hydrous minerals as well as fluids from depth (Rice, 1992) as fault planes are the preferential pathways taken by a degassing earth. Indeed, magnetotelluric measurements along the San Andreas Fault (Becken et al.,

2008) provide additional evidence for fluid channeling within vertical fault zones.

In addition to providing pathways for degassing from below, recent observations and experimental results show that fluid sources can also be generated directly within the fault zone by the earthquake itself. An analysis of pseudotachylytes from the exhumed Nojima fault zone in Japan showed that earthquakes instantaneously expel all CO_2 from the wall rock (Famin, Nakashima, Boullier, Fujimoto, & Hirono, 2008). The earthquake-generated CO_2 would be highly over-pressured, thus contributing to aftershocks and post-seismic slip along the fault plane.

Slip velocity during an earthquake can approach 10 m/s, and recent experiments at high slip velocity show a dramatic weakening (Di et al., 2011). Geochemical signatures measured in the laboratory suggest that this weakening is controlled, at least in carbonate rocks, by thermally-induced decarbonization reactions and substantial weakening by transient thermal pressurization (De et al., 2011). Thermal pressurization has long been suspected of playing an important role in earthquake genesis (Garagash & Rudnicki, 2003; Mase & Smith, 1984, 1987; Rice, 2006), but those studies focussed on fluids already within the fault zone, and did not consider the role of internally created fluids during slip. Future experimental and other studies of frictionally-induced degassing within fault zones will provide constraints on this possibly important mechanism.

4.5 Documented Fluid-Driven Earthquake Sequences

Earthquakes in the Apennines support the notion that deeply trapped CO_2 reservoirs substantially contribute to seismicity, earthquakes, and aftershocks (Chiarabba et al., 2009; Lucente et al., 2010; Miller et al., 2004; Terakawa, Zoporowski, Galvan, & Miller, 2010). The Italian case is especially strong because extensive isotopic studies (Chiodini et al., 2004, 2011) show that the CO_2 is of mantle origin, requiring transport towards the surface that is hydraulically inhibited at depth (Frezzotti, Peccerillo, & Panza, 2009). The conceptual model proposed by Chiodini (Chiodini et al., 2004), and supported in a numerical study (Miller et al., 2004) suggested a dominant role of CO_2 degassing in the 1997 Colfiorito earthquake sequence, while deeply derived CO_2 input into nearby aquifers following the Mw = 6.3 L'Aquila earthquake (Chiodini et al., 2011) lends further support to that scenario. Modeling of the Colfiorito sequence with a non-linear diffusion model (see Figure 1.11) showed that the best fit to the aftershock hypocenters was accomplished using permeability ranging from 10^{-10} to 10^{-13} m^2, indicative

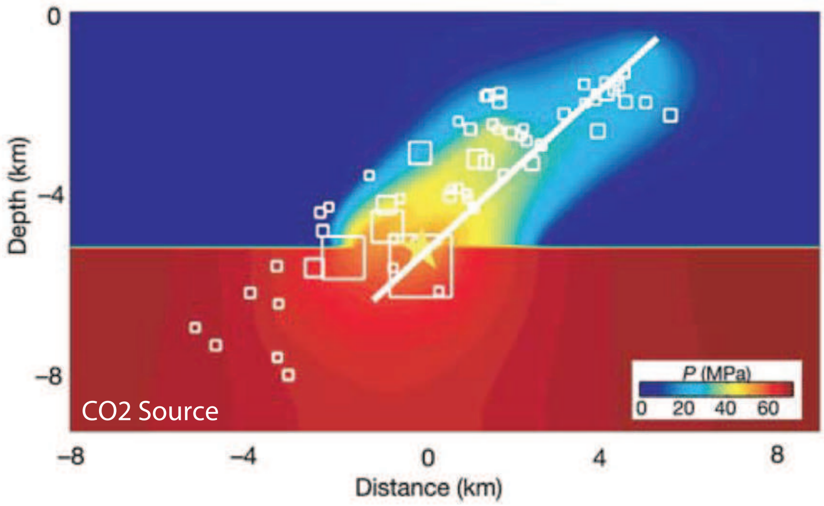


Figure 1.11 Comparison of earthquake hypocenters and calculated fluid pressure field resulting from the co-seismic release of a trapped, high pressure CO_2 reservoir for the 1997 Colfiorito, Italy earthquake sequence. Reprinted from (Miller et al., 2004) with permission of Macmillan Publishers Ltd: Nature. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this book.)

of fracture driven fluid flow and consistent with a growing body of evidence for this range in active tectonic systems (Fisher, 1998; Crone et al., 2011; Waldhauser et al., 2012).

But are the Apennines an exception or the rule? Other earthquakes substantially affected by high pressure fluids at depth include the 2004 Mw = 6.8 Niigata, Japan sequence (Sibson, 2007), northeast Honshu (Sibson, 2009), diffusion-controlled seismicity in Afar (Noir et al., 1997), massive swarms of the Matsushira sequence (Cappa, 2009), the 2011 Mw = 9.0 Tohoku, Japan earthquake (Wang et al., 2012), and an aftershock sequence following the 2004 Mw = 9.2 Sumatra earthquake (Waldhauser et al., 2012). Additionally, trapped fluids at hypocentral depths were inferred from tomographic studies of the hypocenter of the 1995 Kobe, Japan earthquake (Zhao, Kanamori, Negishi, & Wiens, 1996), while mantle-derived CO_2 is implicated for the extended aftershock sequence of the 2001 Mw = 7.9 Bhuj, India earthquake (Mandal, 2011). A fluid influence was also proposed for delaying the Parkfield earthquake (Miller, 1996) expected in the early 1990s but not arriving until 2004. Other evidence for fault zones as principle transportation lanes for deep fluids includes large-scale travertine filled fault zones in Turkey (Uysal et al., 2009), and elevated mantle helium (Newton, 1990).

Meanwhile, indications of latent magmatism was found as an influential trigger in the 2000 Western Tottori, Japan earthquake (Umeda, Asamori, Negi, & Kusano, 2011). Post-seismic rebound attributed to pore-elastic fluid flow has been reported following the Landers earthquake (Peltzer, Rosen, Rogez, & Hudnut, 1996) and in Iceland (Jonsson, Segall, Pedersen, & Bjornsson, 2003), providing additional evidence of fluids trapped at depth.

4.6 Earthquake Swarms

Earthquake swarms are earthquake sequences without a discernible mainshock. Swarms can last weeks and produce many thousands of earthquakes within a relatively small volume. Swarms are observed in volcanic environments, hydrothermal systems, and other active geothermal areas. Fluids are certainly involved in these sequences, but how they appear and dissipate is not certain. The conceptual model proposed by Hill (1977) remains relevant. In this model, a complex mesh of fluid-filled fractures loads high strength bridges isolated from the hydraulic network, producing many small events but without a coherent through-going fracture sufficient to produce one big event. Swarm-type behavior is also found in "enhanced geothermal systems" (EGS), where high pressure fluids are injected into competent rock to stimulate permeability for later exploitation of geothermal resources. This suggests that natural swarm seismicity also involves stimulation of overpressured reservoirs that propagate through Hill-like meshes to generate seismic activity. CO_2 driven swarm seismicity is implicated in many regions, including Bohemia in the Czech Republic (Fischer & Horalek, 2003), in migrating swarms in the lower crust at Mammoth Mountain (Shelly & Hill, 2011), diffusion-like behavior of swarms in the Salton Trough (Chen, Shearer, & Abercrombie, 2012), and in the East African Rift (Lindenfeld, Ruempker, Klemens, Koehn, & Batte, 2012).

A particularly convincing link between CO_2 degassing and seismic swarms was captured in tree rings at Yellowstone (Evans, Bergfeld, McGeehin, King, & Heasler, 2010). Pulses of magmatic CO_2, representing a fivefold increase in CO_2, travelled rapidly through 5 km of brittle crust, indicating a large and transient permeability increase. Another striking case is a 4-m dike injection in the volcanic region of Saudi Arabia over a period of two weeks (Pallister et al., 2010). The dike injection produced thousands of earthquakes as the magma intruded from about 8 km depth to about 2 km depth. The sequence ended when twin normal faults abutting the intrusion pinched shut the hydraulically conductive network.

A case for a purely fault-creep mechanism for swarms has been made for a number of documented swarm sequences along transform faults (Lohman & McGuire, 2007; Roland & McGuire, 2009). In that work, a fluid-driven mechanism was discounted because migration velocities on the order of 0.1–1 km h^{-1} appeared to contradict inferred diffusivities from fluid injection experiments that were modeled as a linear diffusion process (Parotidis, Shapiro, & Rothert, 2005). However, linear diffusion models ignore the significant permeability changes that occur at the local scale of seismic slip, or the cumulative effect of large-scale permeability increase resulting from numerous seismic events in a swarm. Since permeability likely increases by orders of magnitude, the linear diffusion premise is violated.

Swarms are documented around the circum-Pacific (Holtkamp & Brudzinski, 2011), and have been related to slow-slip events in Hawaii (Segall, Desmarais, Shelly, Miklius, & Cervelli, 2006) and a suspected slow slip events in Japan (Okutani & Ide, 2011), so clearly deformation and swarms are related. One way to reconcile slip on transform faults with mobilization of fluids was proposed to explain the coupling between slip and fluid flow during the LUSI mud volcano eruption in East Java, Indonesia (Mazzini et al., 2009). In both scale-model experiments and numerical modeling, Mazzini et al. showed that slip on a transform fault reduces the effective viscosity of the entraining solid, mobilizing buoyant, trapped high-pressure fluids that develop piercing structures co-linear with the strike of the fault. That model showed that slip and fluid-flow are coupled, with slip triggering the flow.

4.7 Triggered Earthquakes

The first unambiguous evidence for distant triggering of earthquakes was found in Long Valley, CA in response to the 1992 Mw = 7.3 Landers earthquake (Hill, Johnston, Langbein, & Bilham, 1995). Perhaps not surprisingly, the Long Valley caldera is a volcanic environment flush with magmatic CO_2. Subsequent studies revealed that earthquake triggering from distant earthquakes is almost invariably associated with hydrothermal systems or volcanic provinces. Examples include the triggering of Yellowstone by the 2002 Mw = 7.9 Denali, Alaska earthquake (Brodsky & Prejean, 2005; Husen, Taylor, Smith, & Healser, 2004; Prejean et al., 2004), triggering of the Cosos geothermal field by the 2010 Mw = 8.8 Maule, Chile earthquake (Peng & Gomberg, 2010), and Love-wave triggered tremor bursts near Parkfield and Cholame along the SAF (Peng & Gomberg, 2010).

The Mw = 9.1 Sumatra earthquake triggered seismicity in southwestern China, including a magnitude 4.6 earthquake coincident with the arrival of the first Rayleigh wave. Seismicity continued during the succeeding two weeks, concentrating near dilatant jogs and fault bends (Lei, 2011). High Vp/Vs ratios in the middle to lower crust, and intense hydrothermal activity, again suggests fluid-induced swarm-like behavior, with dilatant jogs providing extensional kinematics that favor vertical fluid migration.

5. ZEN TREES AND AFTERSHOCKS

The fate of fluids brought to depth is uncertain, but what is certain is that there are only two options; either fluids stay at depth (as free fluid or trapped in hydrous minerals), or they return to the surface. Fluids that make it to shallower levels must do so in an overpressured state because continuous hydraulic connectivity to the free surface is unlikely. According to Byerlee's Law, the strength of the brittle crust is substantial at the base of the seismogenic zone, so if high pressure fluids are trapped they will flow, or pressure will propagate, at rates limited by the permeability. At the base of the seismogenic zone, permeability on the order of 10^{-17}–10^{-19} m^2 is likely, so diffusion processes are slow.

Aftershocks have been discounted as a fluid diffusion process because there appears to be no convincing data showing a $\sqrt{time}$ migration as would be predicted by a diffusion equation. However, pure pressure diffusion is not the right conceptual model from which to start. In my view, aftershock patterns should be more similar to branches of a tree, fed from the same primary stem. In this analogy, the stem feeding the branches is the primary pipe tapping the trapped high pressure reservoir, while the branches are fluid pathways bifurcating from the parent stem. In this conceptual model, an aftershock occurring at location x, y, z at some time t may very well be hydraulically connected through the branch network to another aftershock occurring at the same time t but very far from the point x, y, and z.

If high pressure reservoirs are trapped at the base and continually fed then, late in the seismic cycle, fluid-rock interactions may become increasingly important and contribute to the breakdown process at the earthquake nucleation site. To help visualize how trapped of fluids escape, Figure 1.12 shows a snapshot of a small-scale laboratory experiment where gas at a constant flux is injected into a cohesive clay layer (Dysthe and Svenson, PGP-Oslo unpublished, 2004). The resulting degassing pathway shows a complex topology reflecting the tortuous path taken by the gas. The gas creates its own pathway, sometimes changing directions, making new pathways, and pulsing as pathways are temporarily shut down, reopened, and

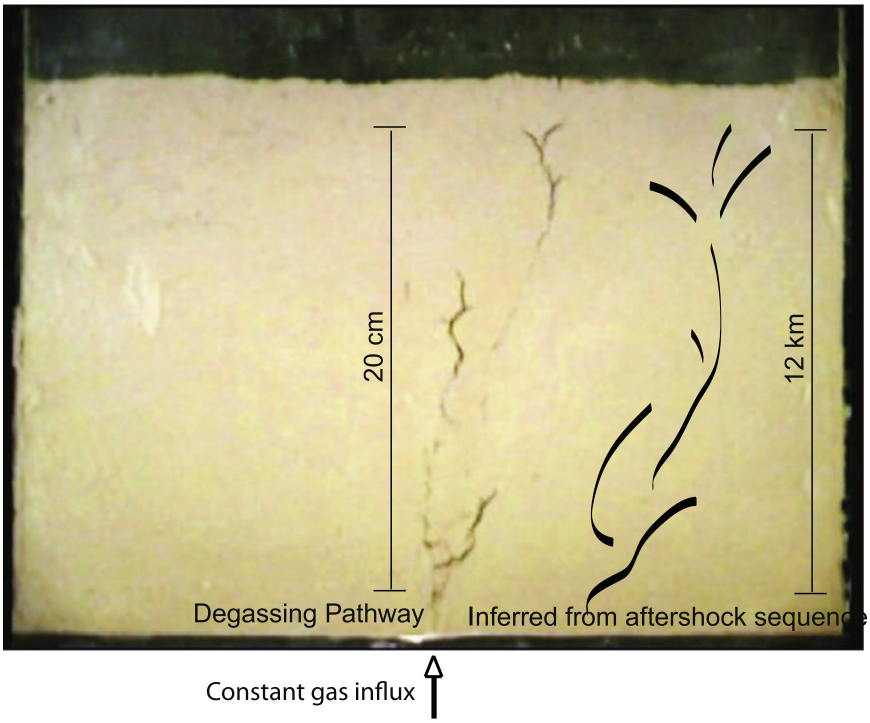


Figure 1.12 Comparison of snapshot of degassing pathway for a high-pressure gas injected into a cohesive clay and the pattern inferred from about 70 sequential aftershocks in the main central step-over of the 1992 Landers earthquake. Scale of the clay experiment is about 20 cm and the scale of the aftershock sequence is about 12 km. Clay experiment result courtesy of Dysthe and Svenson, PGP-Oslo. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this book.)

ever-changing. The pattern at the 0.2 m scale is strikingly similar to that of an aftershock pattern discerned from 70 consecutive aftershocks at a scale of 12,000 m in the main step-over of the 1992 Landers earthquake (Figure 1.12). The striking similarity in pattern suggests that both the clay experiment and the aftershocks are being driven by the same process. Namely, both reflect the degassing pathway and an invasion percolation process (Micklethwaite & Cox, 2004) for an overpressured fluid source at depth.

I call the aftershock patterns, like that in Figure 1.12 "Zen Trees" because they are aesthetically pleasing and reminiscent of Eastern calligraphy. Zen Trees are created by drawing the minimum number of lines that connect all (or most) of the hypocenters from a consecutive sequence of aftershocks

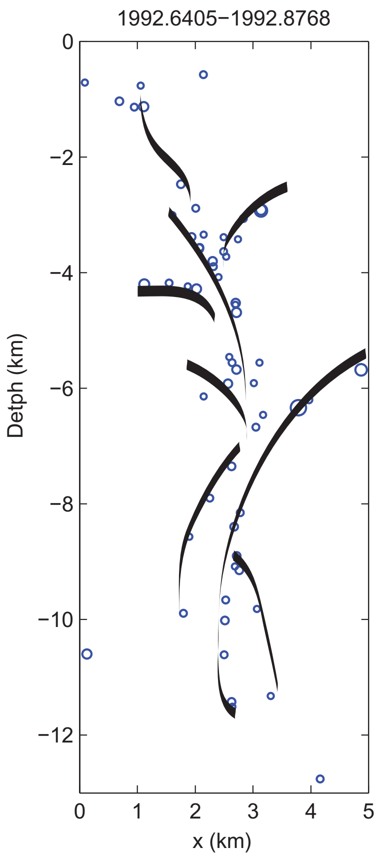


Figure 1.13 Consecutive aftershocks in the main step-over of the Landers earthquake, and interpretive lines drawn through them resulting in a Zen Tree. The time window, in years, is shown at the top of the panel. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this book.)

(Figure 1.13). The resulting image resembles that which would be expected for a degassing pathway. This backbone structure is shown to be stable throughout the entire aftershock sequence, but the shape of the Zen Tree changes through time. This is illustrated in Figure 1.14, where, using the best constrained dataset of hypocenters with relative location errors on the order of tens of meters (Hauksson, Yang, & Shearer, 2012; Lin, Shearer, & Hauksson, 2007), the entire dataset of $M \geqslant 2$ in the central step-over of the Lander's earthquake is parsed into timeframes spanning the catalog (1992–2006). Figure 1.14 shows that the clustered amorphous shape of all events (panel a) actually separates nicely into individual branching structures (panels

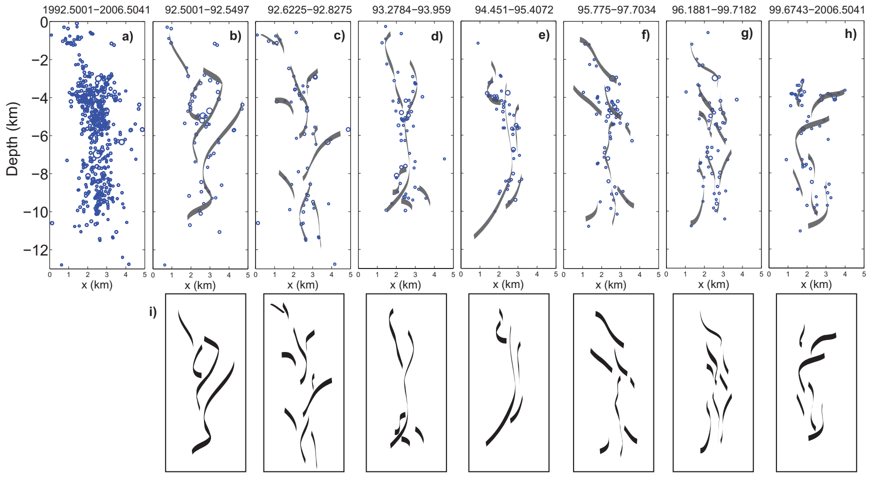


Figure 1.14 (a) Cross-section EW showing the hypocenters of the catalog of all $M > 2$ aftershocks in the central step-over the 1992 Mw = 7.3 Landers, CA earthquake (Lin et al., 2007; Hauksson et al., 2012). (b)–(h) The same data as in (a) but parsed into the timeframes listed above each panel. The Zen Tree interpretation of the data is shown in (i). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this book.)

b–h), producing the Zen Tree catalog for this sequence (panel i). Of course other artistic interpretations are possible with the same dataset, but using the constraint of applying the minimum number of strokes to connect all of the data, any interpretation will produce a Zen Tree.

These structures, appearing in many aftershock sequences in a range of tectonic environments, become clearer with increased hypocenter location accuracy and catalog completeness. Quantifying Zen Trees remains a challenge.

6. NUMERICAL MODELING OF FLUID-ROCK INTERACTIONS

Understanding the role of fluids in tectonic and earthquake processes will necessarily involve numerical modeling to track the complex interactions and feedbacks between deformation, fracturing, fluid flow and chemical reactions. Many numerical models exist, see Rutqvist (2011) for a summary of the state of the art, but the high resolution simulations needed to investigate these systems are difficult to achieve because of the massive

CPU requirements for studies of the underlying physical processes operating at different timescales. A potentially dramatic advance in numerical modeling is the development of codes designed for the Graphical Processing Unit (GPU) architecture. The GPU platform allows fast and high-resolution simulations because of its inherent parallel architecture, and inversion of large matrices is not required if the governing equations are approximated using explicit finite difference algorithms, which are currently the most efficient algorithms on GPU. Developing implicit schemes to run efficiently on the GPU is a current challenge, but advances are being made with potentially

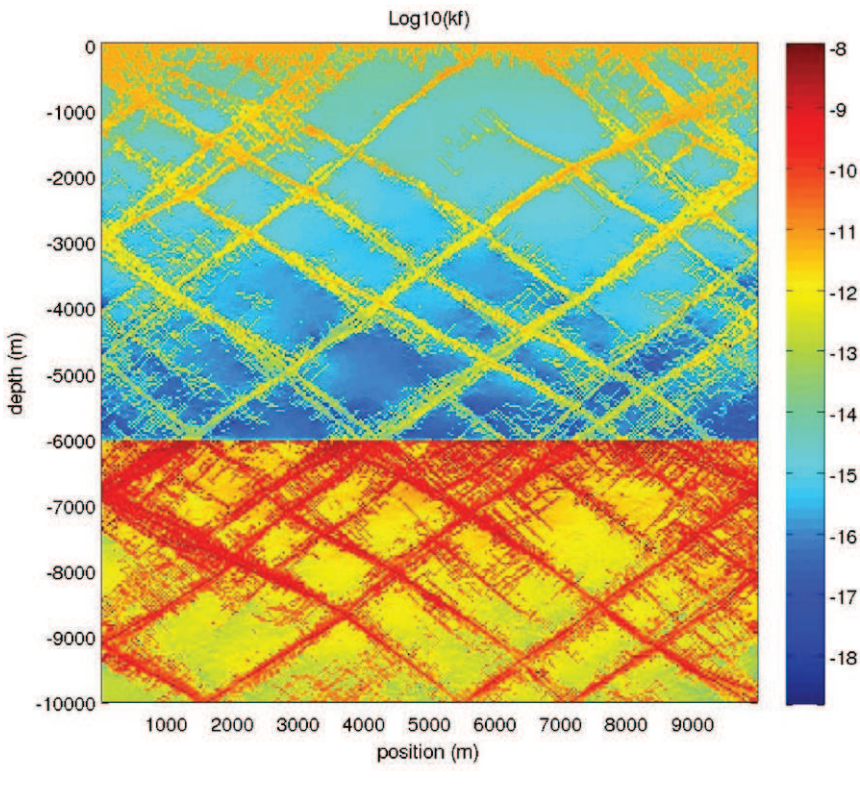


Figure 1.15 Evolved permeability structure in a numerical experiment of compression applied to the side boundaries of pore-elastic plastic rheology coupled with non-linear diffusion and computed on the GPU architecture (Galvan & Miller, 2012). This model uses the FLAC algorithm and numerical fractures grow in response to the far-field boundary conditions and stress interactions from the evolving crack network and poro-elastic stresses. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this book.)

dramatic increases in model resolution and computational speed. As an example, Figure 1.15 shows results of a high resolution simulation (Galvan & MIller, 2012) of permeability evolution within a numerical sample under horizontal compression. The rheology of the model is poro-elastic plastic and coupled to a non-linear diffusion model for fluid pressure, where the non-linearity is introduced by an effective stress-dependent permeability (Rice, 1992; Miller et al., 2004). In this simulation a high pressure source of fluid is trapped the lower part of the model domain, and numerical fractures evolve in response to the far field loading and from stress perturbations from interacting cracks and poro-elastic stresses. This simulation is about an order of magnitude faster using GPU than CPU, indicating great potential for modeling the myriad of processes associated with fluid-rock interaction, deformation, fracture evolution, and transient permeabe networks.

7. CONCLUDING REMARK

The fate of fluids and the tectonic cycle cannot be separated. The enormous volumes of H_2O and CO_2 trapped in the lithosphere must eventually return, and the processes involved in that recycling are complex, transient, and difficult to constrain. From ingestion at mid-ocean ridges to expulsion at plate boundaries and volcanic arcs, high pressure fluids, partially contained in low-permeability environments, can reside for long durations within the lithosphere or within compacting sedimentary basins. However, this low permeability can instantly change to extremely high permeability over timescales of earthquake rupture or dehydration nucleation, and this change dramatically influences the hydro-mechanical-chemical and thermal properties of the system. Future research, constrained by seismological, geochemical, geodetic, and other observations, augmented with more powerful computational platforms and algorithms, will certainly lead to better quantitative understanding of the hydrogeological cycle of the solid earth.

ACKNOWLEDGMENTS

The substantive and careful review of this paper by S.E. Ingebritsen is gratefully acknowledged, and comments by two anonymous reviewers and the editor significantly improved this manuscript. This paper benefitted substantially from many stimulating discussions with J.A.D. Connolly and Y.Y. Podladchikov. This Study was supported in part by a grants from the Volkswagen Foundation and DFG Project MI 1237/2-1.

REFERENCES

Ammon, C. J., Kanamori, H., & Lay, T. (2008). A great earthquake doublet and seismic stress transfer cycle in the central Kuril Islands. *Nature, 451*(7178), 561–565.

Bak, P., Tang, C., & Wiesenfeld, K. (1987). Self-organized criticality—An explanation of 1/f noise. *Physical Review Letters, 59*(4), 381–384.

Beavan, J., Wang, X., Holden, C., Wilson, K., Power, W., Prasetya, G., et al. (2010). Near-simultaneous great earthquakes at Tongan megathrust and outer rise in September 2009. *Nature, 466*, 959–963.

Becken, M., Ritter, O., Bedrosian, P. A., & Weckmann, U. (2011). Correlation between deep fluids, tremor and creep along the central San Andreas fault. *Nature, 480*(7375), 87–90.

Becken, M., Ritter, O., Park, S. K., Bedrosian, P.A., Weckmann, U., & Weber, M. (2008). A deep crustal fluid channel into the San Andreas Fault system near Parkfield, California. *Geophysical Journal International, 173*(2), 718–732.

Beroza, G. C., & Ide, S. (2011). Slow earthquakes and nonvolcanic tremor. *Annual Review of Earth and Planetary Sciences, 39*, 271–296.

Bodnar, R. J., Azbej, T., Becker, S., Cannatelli, C., Fall A., & Severs, M. (2013). Whole Earth geohydrologic cycle: From the clouds to the Core—The role of water in the dynamic Earth System. In: M. E. Bickford (Eds.), *The Web of Geological Sciences: Advances, Impacts, and Interactions.* Geological Society of America Special Paper 500.

Bohnenstiehl, D. R., Waldhauser, F., & Tolstoy, M. (2008). Frequency-magnitude distribution of microearthquakes beneath the 9°50 N region of the East Pacific Rise, October 2003 through April 2004. *Geochemistry Geophysics Geosystems, 9*. doi: http://dx.doi.org/10.1029/2009GC002561.

Brantut, N., Sulem, J., & Schubnel, A. (2011). Effect of dehydration reactions on earthquake nucleation: Stable sliding, slow transients, and unstable slip. *Journal of Geophysical Research, 116*. doi: http://dx.doi.org/10.1029/2010JB007876.

Brodsky, E. E., & Prejean, S. G. (2005). New constraints on mechanisms of remotely triggered seismicity at Long Valley Caldera. *Journal of Geophysical Research, 110*(B4). doi: http://dx.doi.org/10.1029/2004JB003211.

Burov, E. B., & Diament, M. (1995). The effective elastic thickness (T-e) of continental lithosphere—What does it really mean. *Journal of Geophysical Research-Solid Earth, 100*(B3), 3905–3927.

Byerlee, J. (1978). Friction of rocks. *Pure and Applied Geophysics, 116*(4–5), 615–626.

Byerlee, J. (1993). Model for episodic flow of high-pressure water in fault zones before earthquakes. *Geology, 21*(4), 303–306.

Cappa, F. (2009). Modelling fluid transfer and slip in a fault zone when integrating heterogeneous hydrome chanical characteristics in its internal structure. *Geophysical Journal International, 178*(3), 1357–1362.

Carson, B., & Screaton, E. J. (1998). Fluid flow in accretionary prisms: Evidence for focused, time-variable discharge. *Reviews of Geophysics, 36*(3), 329–351.

Chen, X., Shearer, P. M., & Abercrombie, R. E. (2012). Spatial migration of earthquakes within seismic clusters in southern california: Evidence for fluid diffusion. *Journal of Geophysical Research-Solid Earth, 117.*

Chiarabba, C., Amato, A., Anselmi, M., Baccheschi, P., Bianchi, I., Cattaneo, M., et al. (2009). The 2009 L'Aquila (central Italy) Mw 6.3 earthquake: Main shock and aftershocks. *Geophysical Research Letters, 36.* doi: http://dx.doi.org/10.1029/2009GL039627.

Chiodini, G., Caliro, A., Cardellini, C., Frondini, F., Inguaggiato, S., & Matteucci, F. (2011). Geochemical evidence for and characterization of CO_2 rich gas sources in the epicentral area of the Abruzzo 2009 earthquakes. *Earth and Planetary Science Letters, 304*(3–4), 389–398.

Chiodini, G., Cardellini, C., Amato, A., Boschi, E., Caliro, S., Frondini, F., et al. (2004). Carbon dioxide earth degassing and seismogenesis in central and southern Italy. *Geophysical Research Letters, 31.* doi: http://dx.doi.org/10.1029/2004GL019480.

Connolly, J. A. D. (1997). Devolatilization-generated fluid pressure and deformation-propagated fluid flow during prograde regional metamorphism. *Geophysical Research Letters, 102*(B8), 18149–18173.
Connolly, J. A. D. (2010). The mechanics of metamorphic fluid expulsion. *Elements, 6*(3), 165–172.
Connolly, J. A. D., & Podladchikov, Y. Y. (2004). Fluid flow in compressive tectonic settings: Implications for midcrustal seismic reflectors and downward fluid migration. *Journal of Geophysical Research, 109*. doi: http://dx.doi.org/10.1029/2003JB002822.
Connolly, J. A. D., & Podladchikov, Y. Y. (2007). Decompaction weakening and channeling instability in ductile porous media: Implications for asthenospheric melt segregation. *Journal of Geophysical Research, 112*(B10). doi: http://dx.doi.org/10.1029/2005JB004213.
Crone, T. J., Tolstoy, M., & Stroup, D. F. (2011). Permeability structure of young ocean crust from poroelastically triggered earthquakes. *Geophysical Research Letters, 38*. doi: http://dx.doi.org/10.1029/2011GL046820.
De Paola, N., Hirose, T., Mitchell, T., Di Toro, G., Viti, C., & Shimamoto, T. (2011). Fault lubrication and earthquake propagation in thermally unstable rocks. *Geology, 39*(1), 35–38.
Dieterich, J. H. (1979). Modeling of rock friction. 1. Experimental results and constitutive equations. *Journal of Geophysical Research, 84*, 2161–2168.
Dieterich, J. H. (1992). Earthquake nucleation on faults with rate-dependent and state-dependent strength. *Tectonophysics, 211*(1–4), 115–134.
Di Toro, G., Han, R., Hirose, T., De Paola, N., Nielsen, S., Mizoguchi, K., et al. (2011). Fault lubrication during earthquakes. *Nature, 471*(7339), 494–498. doi: http://dx.doi.org/10.1038/nature09838.
Duretz, T., Gerya, T. V., & May, D. A. (2011). Numerical modelling of spontaneous slab breakoff and subsequent topographic response. *Tectonophysics, 502*(1–2), 244–256.
England, P. C., & Katz, R. F. (2010). Global systematics of arc volcano position. *Nature, 468*(7325), E6–E7.
Evans, W. C., Bergfeld, D., McGeehin, J. P., King, J. C., & Heasler, H. (2010). Tree-ring (14)C links seismic swarm to CO_2 spike at Yellowstone, USA. *Geology, 38*(12), 1075–1078.
Faccenda, M., Gerya, T. V., & Burlini, L. (2009). Deep slab hydration induced by bending-related variations in tectonic pressure. *Nature Geoscience, 2*(11), 790–793.
Faccenda, M., Gerya, T. V., Mancktelow, N. S., & Moresi, L. (2012). Fluid flow during slab unbending and dehydration: Implications for intermediate-depth seismicity, slab weakening and deep water recycling. *Geochemistry Geophysics Geosystems, 13*. doi: http://dx.doi.org/10.1029/2011GC003860.
Famin, V., Nakashima, S., Boullier, A. M., Fujimoto, K., & Hirono, T. (2008). Earthquakes produce carbon dioxide in crustal faults. *Earth and Planetary Science Letters, 265*(3–4), 487–497.
Faulkner, D. R., Mitchell, T. M., Behnsen, J., Hirose, T., & Shimamoto, T. (2011). Stuck in the mud? Earthquake nucleation and propagation through accretionary forearcs. *Geophysical Research Letters, 38*. doi: http://dx.doi.org/10.1029/2011GL048552.
Fischer, T., & Horalek, J. (2003). Space-time distribution of earthquake swarms in the principal focal zone of the NW Bohemia/Vogtland seismoactive region: Period 1985–2001. *Journal of Geodynamics, 35*(1–2), 125–144.
Fisher, A. T. (1998). Permeability within basaltic oceanic crust. *Reviews of Geophysics, 36*(2), 143–182.
Frezzotti, M. L., Peccerillo, A., & Panza, G. (2009). Carbonate metasomatism and CO_2 lithosphere asthenosphere degassing beneath the Western Mediterranean: An integrated model arising from petrological and geophysical data. *Chemical Geology, 262*(1–2), 108–120.
Galvan, B., & Miller, S. (2012). A full GPU simulation of evolving fracture networks in a heterogeneous poro-elasto-plastic medium with effective-stress-dependent permeability.

In D. Yuen (Ed.), *Proceedings of the first conference on GPU solutions to multi-scale problems in science and engineering*, Harbin, China.

Garagash, D. I., & Rudnicki, J. W. (2003). Shear heating of a fluid-saturated slip-weakening dilatant fault zone—1. Limiting regimes. *Journal of Geophysical Research, 108*(B2). doi: http://dx.doi.org/10.1029/2001JB001653.

Gerya, T. (2011). Future directions in subduction modeling. *Journal of Geodynamics, 52*(5), 344–378.

Gerya, T. V., Yuen, D. A., & Maresch, W. V. (2004). Thermomechanical modelling of slab detachment. *Earth and Planetary Science Letters, 226*(1–2), 101–116.

Gomberg, J. (2010). Introduction to special section on phenomenology, underlying processes, and hazard implications of aseismic slip and nonvolcanic tremor. *Journal of Geophysical Research, 115*. doi: http://dx.doi.org/10.1029/2010JB008052.

Gomberg, J., & Cascadia Working Group (2010). Slow-slip phenomena in Cascadia from 2007 and beyond: A review. *Geological Society of America Bulletin, 122*(7–8), 963–978.

Gorman, P. J., Kerrick, D. M., & Connolly, J. A. D. (2006). Modeling open system metamorphic decarbonation of subducting slabs. *Geochemistry Geophysics Geosystems, 7*. doi: http://dx.doi.org/10.1029/2005GC001125.

Grassi, D., & Schmidt, M. W. (2011). The melting of carbonated pelites from 70 to 700 km depth. *Journal of Petrology, 52*(4), 765–789.

Gratier, J.-P., Richard, J., Renard, F., Mittempergher, S., Doan, M. L., Di Toro, G., Hadizadeh, J., Boullier, A.-M. (2011). Aseismic sliding of active faults by pressure solution creep: Evidence from the San Andreas fault observatory at depth. *Geology, 39*, 1131–1134.

Grove, T. L., Till, C. B., Lev, E., Chatterjee, N., & Medard, E. (2009). Kinematic variables and water transport control the formation and location of arc volcanoes. *Nature, 459*(7247), 694–697.

Hammouda, T. (2003). High-pressure melting of carbonated eclogite and experimental constraints on carbon recycling and storage in the mantle. *Earth and Planetary Science Letters, 214*(1–2), 357–368.

Hauksson, E., Yang, W., & Shearer, P. M. (2012). Waveform relocated earthquake catalog for southern California (1981 to 2011). *Bulletin of the Seismological Society of America, 102*, 2239–2244. doi: 10.1785/0120120010.

Hilairet, N., Reynard, B., Wang, Y. B., Daniel, I., Merkel, S., Nishiyama, N., et al. (2007). High-pressure creep of serpentine, interseismic deformation, and initiation of subduction. *Science, 318*(5858), 1910–1913.

Hill, D. P. (1977). Model for earthquake swarms. *Journal of Geophysical Research, 82*(8), 1347–1352.

Hill, D. P., Johnston, M. J. S., Langbein, J. O., & Bilham, R. (1995). Response of long-valley caldera to the Mw = 7.3 landers, California, earthquake. *Journal of Geophysical Research, 100*(B7), 12985–13005.

Hillers, G., Ben-Zion, Y., & Mai, P. M. (2006). Seismicity on a fault controlled by rate- and state-dependent friction with spatial variations of the critical slip distance. *Journal of Geophysical Research-Solid Earth, 111*. doi: http://dx.doi.org/10.1029/2005JB003859.

Hillers, G., & Miller, S. A. (2006). Stability regimes of a dilatant, fluid-infiltrated fault plane in a three dimensional elastic solid. *Journal of Geophysical Research, 111*(B8). doi: http://dx.doi.org/10.1029/2005JB003872.

Hillers, G., & Miller, S. A. (2007). Dilatancy controlled spatiotemporal slip evolution of a sealed fault with spatial variations of the pore pressure. *Geophysical Journal International, 168*(1), 431–445.

Holtkamp, S. G., & Brudzinski, M. R. (2011). Earthquake swarms in circum-pacific subduction zones. *Earth and Planetary Science Letters, 305*(1–2), 215–225.

Hubbert, M. K., & Rubey, W. W. (1959). Role of fluid pressure in mechanics of overthrust faulting. 1. Mechanics of fluid-filled porous solids and its application to overthrust faulting. *Geological Society of America Bulletin, 70*(2), 115–166.

Husen, S., Taylor, R., Smith, R. B., & Healser, H. (2004). Changes in geyser eruption behavior and remotely triggered seismicity in Yellowstone National Park produced by the 2002 M 7.9 Denali fault earthquake, Alaska. *Geology, 32*(6), 537–540.

Hyndman, R. D., & Peacock, S. M. (2003). Serpentinization of the forearc mantle. *Earth and Planetary Science Letters, 212*(3–4), 417–432.

Ide, S. (2008). A Brownian walk model for slow earthquakes. *Geophysical Research Letters, 35*(17). doi: http://dx.doi.org/10.1029/2008GL034821.

Ingebritsen, S. E., & Manning, C. E. (2002). Diffuse fluid flux through orogenic belts: Implications for the world ocean. *Proceedings of the National Academy of Sciences of the United States of America, 99*(14), 9113–9116.

Ingebritsen, S. E., & Manning, C. E. (2010). Permeability of the continental crust: Dynamic variations inferred from seismicity and metamorphism. *Geofluids, 10*(1–2), 193–205.

Irwin, W. P., & Barnes, I. (1980). Tectonic relations of carbon-dioxide discharges and earthquakes. *Journal of Geophysical Research, 85*(NB6), 3115–3121.

Iwamori, H. (2007). Transportation of H_2O beneath the Japan arcs and its implications for global water circulation. *Chemical Geology, 239*(3–4), 182–198.

Jamtveit, B. (2010). Metamorphism: From patterns to processes. *Elements, 6*(3), 149–152.

Jonsson, S., Segall, P., Pedersen, R., & Bjornsson, G. (2003). Post-earthquake ground movements correlated to pore-pressure transients. *Nature, 424*(6945), 179–183.

Kanamori, H., & Kikuchi, M. (1993). The 1992 Nicaragua earthquake—A slow tsunami earthquake associated with subducted sediments. *Nature, 361*(6414), 714–716.

Kato, A., Obara, K., Igarashi, T., Tsuruoka, H., Nakagawa, S., & Hirata, N. (2012). Propagation of slow slip leading up to the 2011 Mw 9.0 Tohoku-Oki earthquakes. *Science, 335*(6069), 705–708.

Kawakatsu, H., & Watada, S. (2007). Seismic evidence for deep-water transportation in the mantle. *Science, 316*(5830), 1468–1471.

Kelemen, P. B., Matter, J., Streit, E. E., Rudge, J. F., Curry, W. B., & Blusztajn, J. (2011). Rates and mechanisms of mineral carbonation in peridotite: Natural processes and recipes for enhanced, in situ CO_2 capture and storage. *Annual Review of Earth and Planetary Sciences, 39*, 545–576.

Kerrick, D. M. (2001). Present and past nonanthropogenic CO_2 degassing from the solid Earth. *Reviews of Geophysics, 39*(4), 565–585.

Kerrick, D. M., & Connolly, J. A. D. (2001a). Metamorphic devolatilization of subducted marine sediments and the transport of volatiles into the Earth's mantle. *Nature, 411*(6835), 293–296.

Kerrick, D.M., & Connolly, J.A.D. (2001b). Metamorphic devolatilization of subducted oceanic metabasalts: Implications for seismicity, arc magmatism and volatile recycling. *Earth and Planetary Science Letters, 189*(1–2), 19–29.

Kirby, S. H. (2000). Earth science—Taking the temperature of slabs. *Nature, 403*(6765), 31–34.

Kirby, S. H., Stein, S., Okal, E. A., & Rubie, D. C. (1996). Metastable mantle phase transformations and deep earthquakes in subducting oceanic lithosphere. *Reviews of Geophysics, 34*(2), 261–306.

Lamb, S. (2006). Shear stresses on megathrusts: Implications for mountain building behind subduction zones. *Journal of Geophysical Research, 111*(B7). doi: http://dx.doi.org/10.1029/2005JB003916.

Lei, X. (2011). Possible roles of the Zipingpu Reservoir in triggering the 2008 Wenchuan earthquake. *Journal of Asian Earth Sciences, 40*(4), 844–854.

Lin, G., Shearer, P. M., & Hauksson, E. (2007). Applying a three-dimensional velocity model, waveform cross correlation, and cluster analysis to locate southern California seismicity from 1981 to 2005. *Journal of Geophysical Research, 112*, B12309. doi: http://dx.doi.org/10.1029/2007JB004986.

Lindenfeld, M., Ruempker, G., Klemens, L., Koehn, D., & Batte, A. (2012). Fluid-triggered earthquake swarms in the Rwenzori region, East African Rift: Evidence for rift initiation. *Tectonophysics,* 95–104.

Liu, Y., Rice, J. R., & Larson, K. M. (2007). Seismicity variations associated with aseismic transients in Guerrero, Mexico, 1995–2006. *Earth and Planetary Science Letters, 262*(3–4), 493–504.

Liu, Y. J., & Rice, J. R. (2005). Aseismic slip transients emerge spontaneously in three-dimensional rate and state modeling of subduction earthquake sequences. *Journal of Geophysical Research, 110*(B8). doi: http://dx.doi.org/10.1029/2004JB003424.

Liu, Y. J., & Rubin, A. M. (2010). Role of fault gouge dilatancy on aseismic deformation transients. *Journal of Geophysical Research, 115*. doi: http://dx.doi.org/10.1029/2010JB007522.

Lohman, R. B., & McGuire, J. J. (2007). Earthquake swarms driven by aseismic creep in the Salton Trough, California. *Journal of Geophysical Research, 112*. doi: http://dx.doi.org/10.1029/2006JB004596.

Lucente, F. P., De Gori, P., Margheriti, L., Piccinini, D., Di Bona, M., Chiarabba, C., et al. (2010). Temporal variation of seismic velocity and anisotropy before the 2009 Mw 6.3 L'Aquila earthquake, Italy. *Geology, 38*(11), 1015–1018.

Mandal, P. (2011). Crustal and lithospheric thinning beneath the seismogenic Kachchh rift zone, Gujarat (India): Its implications toward the generation of the 2001 Bhuj earthquake sequence. *Journal of Asian Earth Sciences, 40*(1), 150–161.

Mase, C. W., & Smith, L. (1984). Pore-fluid pressures and frictional heating on a fault surface. *Pure and Applied Geophysics, 122*(2–4), 583–607.

Mase, C. W., & Smith, L. (1987). Effects of frictional heating on the thermal, hydrologic, and mechanical response of a fault. *Journal of Geophysical Research, 92*(B7), 6249–6272.

Mazzini, A., Nermoen, A., Krotkiewski, M., Podladchikov, Y., Planke, S., & Svensen, H. (2009). Strike-slip faulting as a trigger mechanism for overpressure release through piercement structures. Implications for the LUSI mud volcano, Indonesia. *Marine and Petroleum Geology, 26*(9), 1751–1765.

McLean, D. M. (1985). Deccan traps mantle degassing in the terminal Cretaceous marine extinctions. *Cretaceous Research, 6*(3), 235–259.

Micklethwaite, S., & Cox, S. F. (2004). Fault-segment rupture, aftershock-zone fluid flow, and mineralization. *Geology, 32*(9), 813–816.

Miller, S. A. (1996). Fluid-mediated influence of adjacent thrusting on the seismic cycle at Parkfield. *Nature, 382*(6594), 799–802.

Miller, S. A. (2002). Properties of large ruptures and the dynamical influence of fluids on earthquakes and faulting. *Journal of Geophysical Research, 107*. doi: http://dx.doi.org/10.1029/2000JB000032.

Miller, S. A., Ben-Zion, Y., & Burg, J. P. (1999). A three-dimensional fluid-controlled earthquake model: Behavior and implications. *Journal of Geophysical Research, 104*(B5), 10621–10638.

Miller, S. A., Collettini, C., Chiaraluce, L., Cocco, M., Barchi, M., & Kaus, B. J. P. (2004). Aftershocks driven by a high-pressure CO_2 source at depths. *Nature, 427*(6976), 724–727.

Miller, S. A., & Nur, A. (2000). Permeability as a toggle switch in fluid-controlled crustal processes. *Earth and Planetary Science Letters, 183*(1–2), 133–146.

Miller, S. A., Nur, A., & Olgaard, D. L. (1996). Earthquakes as a coupled shear stress high pore pressure dynamical system. *Geophysical Research Letters, 23*(2), 197–200.

Miller, S.A., van der Zee, W., Olgaard, D. L., & Connolly, J. A. D. (2003). A fluid-pressure feedback model of dehydration reactions: Experiments, modelling, and application to subduction zones. *Tectonophysics, 370*(1–4), 241–251.
Mitchell, T., & Faulkner, D. R. (2009). The nature and origin of off-fault damage surrounding strike-slip fault zones with a wide range of displacements: A field study from the Atacama fault system, northern Chile. *Journal of Structural Geology, 31*, 802–816.
Mohais, R., Xu, C., Dowd, P. A., & Hand, M. (2012). Permeability correction factor for fractures with permeable walls. *Geophysical Research Letters, 39*. doi: http://dx.doi.org/10.1029/2011GL050519.
Moore, J. C., & Vrolijk, P. (1992). Fluids in accretionary prisms. *Reviews of Geophysics, 30*(2), 113–135.
Moreno, M., Melnick, D., Rosenau, M., Baez, J., Klotz, J., Oncken, O., et al. (2012). Toward understanding tectonic control on the M-w 8.8 2010 Maule Chile earthquakes. *Earth and Planetary Science Letters, 321*, 152–165.
Moreno, M., Rosenau, M., & Oncken, O. (2010). 2010 Maule earthquake slip correlates with pre-seismic locking of Andean subduction zones. *Nature, 467*(7312), 198–U84.
Newton, R. C. (1990). Fluids and shear zones in the deep crust. *Tectonophysics, 182*(1–2), 21–37.
Noir, J., Jacques, E., Bekri, S., Adler, P. M., Tapponnier, P., & King, G. C. P. (1997). Fluid flow triggered migration of events in the 1989 Dobi earthquake sequence of Central Afar. *Geophysical Research Letters, 24*(18), 2335–2338.
Nur, A., & Booker, J. R. (1972). Aftershocks caused by pore fluid flow? *Science, 175*(4024), 885–887.
Nur, A., & Byerlee, J. D. (1971). Exact effective stress law for elastic deformation of rock with fluids. *Journal of Geophysical Research, 76*(26), 6414–6419.
Okutani, T., & Ide, S. (2011). Statistic analysis of swarm activities around the Boso Peninsula, Japan: Slow slip events beneath Tokyo Bay? *Earth Planets and Space, 63*(5), 419–426 (Okutani, Tsubasa Ide, Satoshi).
Pallister, J. S., McCausland, W. A., Jonsson, S., Lu, Z., Zahran, H. M., El Hadidy, S., et al. (2010). Broad accommodation of rift-related extension recorded by dyke intrusion in Saudi Arabia. *Nature Geoscience, 3*(10), 705–712.
Parotidis, M., Shapiro, S. A., & Rothert, E. (2005). Evidence for triggering of the vogtland swarms 2000 by pore pressure diffusion. *Journal of Geophysical Research, 110*(B5).
Peacock, S. M. (2001). Are the lower planes of double seismic zones caused by serpentine dehydration in subducting oceanic mantle? *Geology, 29*(4), 299–302.
Peacock, S. M., Christensen, N. I., Bostock, M. G., & Audet, P. (2011). High pore pressures and porosity at 35 km depth in the Cascadia subduction zones. *Geology, 39*(5), 471–474.
Peacock, S. M., & Wang, K. (1999). Seismic consequences of warm versus cool subduction metamorphism: Examples from southwest and northeast Japan. *Science, 286*(5441), 937–939.
Peltzer, G., Rosen, P., Rogez, F., & Hudnut, K. (1996). Postseismic rebound in fault step-overs caused by pore fluid flow. *Science, 273*(5279), 1202–1204.
Peng, Z. G., & Gomberg, J. (2010). An integrated perspective of the continuum between earthquakes and slow-slip phenomena. *Nature Geoscience, 3*(9), 599–607.
Perfettini, H., & Ampuero, J. P. (2008). Dynamics of a velocity strengthening fault region: Implications for slow earthquakes and postseismic slip. *Journal of Geophysical Research, 113*. doi: http://dx.doi.org/10.1029/2007JB005398.
Poli, S., & Schmidt, M. W. (2002). Petrology of subducted slabs. *Annual Review of Earth and Planetary Sciences, 30*, 207–235.
Prejean, S. G., Hill, D. P., Brodsky, E. E., Hough, S. E., Johnston, M. J. S., Malone, S. D., et al. (2004). Remotely triggered seismicity on the United States west coast following the Mw 7.9 Denali fault earthquakes. *Bulletin of the Seismological Society of America, 94*(6), S348–S359.

Ranero, C. R., Morgan, J. P., McIntosh, K., & Reichert, C. (2003). Bending-related faulting and mantle serpentinization at the middle America trench. *Nature, 425*(6956), 367–373.
Regenauer-Lieb, K., Yuen, D. A., & Branlund, J. (2001). The initiation of subduction: Criticality by addition of water? *Science, 294*(5542), 578–580.
Rice, J. (1992). Fault stress states, pore pressure distributions, and the weakness of the San Andreas faults. In B. Evans & T.-f. Wong (Eds.), *Fault mechanics and the transport properties of rocks* (p. 475–501). Academic Press.
Rice, J. R. (2006). Heating and weakening of faults during earthquake slip. *Journal of Geophysical Research, 111*(B5). doi: http://dx.doi.org/10.1029/2005JB004006.
Richardson, C. N., Lister, J. R., & McKenzie, D. (1996). Melt conduits in a viscous porous matrix. *Journal of Geophysical Research, 101*(B9), 20423–20432.
Rogers, G., & Dragert, H. (2003). Episodic tremor and slip on the Cascadia subduction zone: The chatter of silent slips. *Science, 300*(5627), 1942–1943.
Roland, E., & McGuire, J. J. (2009). Earthquake swarms on transform faults. *Geophysical Journal International, 178*(3), 1677–1690.
Rozhko, A. Y. (2010). Role of seepage forces on seismicity triggering. *Journal of Geophysical Research-Solid Earth, 115*. doi: http://dx.doi.org/10.1029/2009JB007182.
Rubinstein, J. L., Gomberg, J., Vidale, J. E., Wech, A. G., Kao, H., Creager, K. C., et al. (2009). Seismic wave triggering of nonvolcanic tremor, episodic tremor and slip, and earthquakes on vancouver island. *Journal of Geophysical Research, 114*.
Rupke, L. H., Morgan, J. P., Hort, M., & Connolly, J. A. D. (2004). Serpentine and the subduction zone water cycle. *Earth and Planetary Science Letters, 223*(1–2), 17–34.
Rutqvist, J. (2011). Status of the TOUGH-FLAC simulator and recent applications related to coupled fluid flow and crustal deformations. *Computers and Geosciences, 37*(6), 739–750.
Saffer, D. M., & Tobin, H. J. (2011). Hydrogeology and mechanics of subduction zone forearcs: Fluid flow and pore pressure. *Annual Review of Earth and Planetary Sciences, 39*, 157–186.
Scholz, C. H. (1998). Earthquakes and friction laws. *Nature, 391*(6662), 37–42.
Scholz, C. H., & Campos, J. (1995). On the mechanism of seismic decoupling and back-arc spreading at subduction zones. *Journal of Geophysical Research, 100*(B11), 22103–22115.
Scott, D. R., Stevenson, D. J., & Whitehead, J. A. (1986). Observations of solitary waves in a viscously deformable pipe. *Nature, 319*(6056), 759–761.
Segall, P., Desmarais, E. K., Shelly, D., Miklius, A., & Cervelli, P. (2006). Earthquakes triggered by silent slip events on Kilauea volcano, Hawaii. *Nature, 442*(7098), 71–74.
Segall, P., & Rice, J. R. (1995). Dilatancy, compaction, and slip instability of a fluid-infiltrated fault. *Journal of Geophysical Research, 100*(B11), 22155–22171.
Segall, P., Rubin, A. M., Bradley, A. M., & Rice, J. R. (2010). Dilatant strengthening as a mechanism for slow slip events. *Journal of Geophysical Research, 115*. doi: http://dx.doi.org/10.1029/2010JB007449.
Shaw, B. E., & Rice, J. R. (2000). Existence of continuum complexity in the elastodynamics of repeated fault ruptures. *Journal of Geophysical Research, 105*(B10), 23791–23810.
Shelly, D. R., Beroza, G. C., & Ide, S. (2007). Non-volcanic tremor and low-frequency earthquake swarms. *Nature, 446*(7133), 305–307.
Shelly, D. R., Beroza, G. C., Ide, S., & Nakamula, S. (2006). Low-frequency earthquakes in Shikoku, Japan, and their relationship to episodic tremor and slips. *Nature, 442*(7099), 188–191.
Shelly, D. R., & Hill, D. P. (2011). Migrating swarms of brittle-failure earthquakes in the lower crust beneath Mammoth Mountain, California. *Geophysical Research Letters, 38*.
Shibazaki, B. (2005). Nucleation process with dilatant hardening on a fluid-infiltrated strike-slip fault model using a rate- and state-dependent friction law. *Journal of Geophysical Research, 110*(B11).
Shibazaki, B., Garatani, K., & Okuda, H. (2007). Finite element analysis of crustal deformation in the Ou Backbone Range, northeastern Japan, with non-linear visco-elasticity

and plasticity: Effects of non-uniform thermal structures. *Earth Planets and Space, 59*(6), 499–512.
Shibazaki, B., & Iio, Y. (2003). On the physical mechanism of silent slip events along the deeper part of the seismogenic zone. *Geophysical Research Letters, 30*(9).
Sibson, R. H. (1973). Interactions between temperature and pore-fluid pressure during earthquake faulting and a mechanism for partial or total stress relief. *Nature, 243*(126), 66–68.
Sibson, R. H. (1996). Structural permeability of fluid-driven fault-fracture meshes. *Journal of Structural Geology, 18*(8), 1031–1042.
Sibson, R. H. (2000). Fluid involvement in normal faulting. *Journal of Geodynamics, 29*(3–5), 469–499.
Sibson, R. H. (2003). Brittle-failure controls on maximum sustainable overpressure in different tectonic regimes. *AAPG Bulletin, 87*(6), 901–908.
Sibson, R. H. (2007). An episode of fault-valve behaviour during compressional inversion? the 2004 M 6.8 Mid-Niigata Prefecture, Japan, earthquake sequences. *Earth and Planetary Science Letters, 257*(1–2), 188–199.
Sibson, R. H. (2009). Rupturing in overpressured crust during compressional inversion-the case from NE Honshu, Japan. *Tectonophysics, 473*(3–4), 404–416.
Simpson, G. D. H. (2010). Formation of accretionary prisms influenced by sediment subduction and supplied by sediments from adjacent continents. *Geology, 38*(2), 131–134.
Spinelli, G. A., & Wang, K. (2009). Links between fluid circulation, temperature, and metamorphism in subducting slabs. *Geophysical Research Letters, 36*. doi: http://dx.doi.org/10.1029/2009GL038706.
Syracuse, E. M., van Keken, P. E., & Abers, G. A. (2010). The global range of subduction zone thermal models. *Physics of the Earth and Planetary Interiors, 183*(1–2), 73–90.
Terakawa, T., Miller, S. A., & Deichmann, N. (2012). High fluid pressure and triggered earthquakes in the enhanced geothermal system in basel, switzerland. Journal of Geophysical Research-Solid Earth, 117. doi: 10.1029/2011JB008980.
Terakawa, T., Zoporowski, A., Galvan, B., & Miller, S. A. (2010). High pressure fluid at hypocentral depths in the L'Aquila region inferred inferred from earthquake focal mechanisms. *Geology, 38*, 995–998.
Terzaghi, K. (1923). Die Berechnung des Durchlassigkeitsziffer des Tones aus dem Verlauf der hydrodynamischen Spannungserscheinungen. *Sitz. Akad. Wiss. Wien, abt IIa, 132*, 125–138.
Thomas, A. M., Nadeau, R. M., & Burgmann, R. (2009). Tremor-tide correlations and near-lithostatic pore pressure on the deep San Andreas fault. *Nature, 462*(7276), 1048–1051.
Tobin, H. J., Moore, J. C., Mackay, M. E., Orange, D. L., & Kulm, L. D. (1993). Fluid-flow along a strike-slip fault at the toe of the Oregon accretionary prism—Implications for the geometry of frontal accretion. *Geological Society of America Bulletin, 105*(5), 569–582.
Tolstoy, M., Waldhauser, F., Bohnenstiehl, D. R., Weekly, R. T., & Kim, W. Y. (2008). Seismic identification of along-axis hydrothermal flow on the East Pacific Rise. *Nature, 451*(7175), 181–184.
Umeda, K., Asamori, K., Negi, T., & Kusano, T. (2011). A large intraplate earthquake triggered by latent magmatism. *Journal of Geophysical Research, 116*. doi: http://dx.doi.org/10.1029/2010JB007963.
Uysal, I. T., Feng, Y. X., Zhao, J. X., Isik, V., Nuriel, P., & Golding, S. D. (2009). Hydrothermal CO_2 degassing in seismically active zones during the late Quaternary. *Chemical Geology, 265*(3–4), 442–454.
van Keken, P.E., Hacker, B.R., Syracuse, E.M., & Abers, G.A. (2011). Subduction factory: 4. depth-dependent flux of H_2O from subducting slabs worldwide. *Journal of Geophysical Research, 116*. doi: http://dx.doi.org/10.1029/2010JB007922.
van der Meer, D. G., Spakman, W., van Hinsbergen, D. J. J., Amaru, M. L., & Torsvik, T. H. (2010). Towards absolute plate motions constrained by lower-mantle slab remnants. *Nature Geoscience, 3*(1), 36–40.

Vonhuene, R., & Scholl, D. W. (1991). Observations at convergent margins concerning sediment subduction, subduction erosion, and the growth of continental-crust. *Reviews of Geophysics, 29*(3), 279–316.

Walder, J., & Nur, A. (1984). Porosity reduction and crustal pore pressure development. *Journal of Geophysical Research, 89*(NB13), 1539–1548.

Waldhauser, F., Schaff, D. P., Diehl, T., & Engdahl, E. R. (2012). Splay faults imaged by fluid-driven aftershocks of the 2004 Mw 9.2 Sumatra-Andaman earthquake. *Geology, 40*, 243–246. doi: http://dx.doi.org/10.1130/G32420.1.

Wang, Z., Huang, W., Zhao, D., & Pei, S. (2012). Mapping the Tohoku forearc: Implications for the mechanism of the 2011 East Japan earthquake (Mw 9.0). *Tectonophysics*, 147–154.

Wortel, M. J. R., & Spakman, W. (2000). Geophysics—subduction and slab detachment in the Mediterranean-Carpathian region. *Science, 290*(5498), 1910–1917.

Zhang, S. Q., & Cox, S. F. (2000). Enhancement of fluid permeability during shear deformation of a synthetic mud. *Journal of Structural Geology, 22*(10), 1385–1393.

Zhao, D. P., Kanamori, H., Negishi, H., & Wiens, D. (1996). Tomography of the source area of the 1995 Kobe earthquake: Evidence for fluids at the hypocenter? *Science, 274*(5294), 1891–1894.

View publication stats